

INTERVIEW PREPARATION GUIDE

Frontend Developer WeMakeScholars

React · Next.js · Performance & SEO · Machine Coding · HR
Prepared for Sujith Mondithoka · Interview: 10:00 AM

How to use this guide. Work the sections in order. The plan on the next page is timed — if you fall behind, drop from the bottom, never from the top.

● **Red dot** = high priority, almost certainly asked. ●● = your differentiator. ● **Amber** = worth knowing.

✓ = self-check before moving on. ⚠ = common trap or something to verify.

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WeMakeScholars — Frontend Developer Interview Prep

Your plan for tonight and tomorrow morning · Interview: 10:00 AM

| How this guide is written

Every topic follows the same shape, so you always know what you are reading:

1. **The context** — what the thing is, and what problem it solves, in plain words.
2. **A simple example** — small code you can actually follow.
3. **Why it matters** — where it is used in real work.
4. **"Say this"** — the answer to give in the room.

You are not meant to memorise the answers. You are meant to understand the idea, so that when the question comes out slightly differently, you can still answer it.

| 1. Where you actually stand

Read this first, because it changes how you should behave tomorrow.

You are not a fresher. You have about 1 year 4 months of production experience at Zensar building for Standard Bank South Africa, plus a 5 month internship. You have shipped features used by 10,000+ corporate banking clients, and you have real numbers to prove it (Lighthouse 62 to 88, 50,000+ daily audit transactions, 60% faster card processing).

Most people applying for this role will have college projects. You have production React and TypeScript in a bank, plus full stack Spring Boot work.

Three consequences:

1. **Do not undersell yourself.** No "I only did small parts". Speak like someone who has shipped software that handles other people's money.
2. **Aim at the top of the 4 to 6 LPA band**, not the middle. See file 09.
3. **You are an immediate joiner.** Your last role ended in March, so you can start now while most candidates are serving 60 or 90 days. That is worth real money to a hiring manager. Say it clearly.

But your resume also creates four questions you must be ready for: the gap since March, why you left Zensar, why a pure frontend role when you have Java, and how much Next.js you have really used. All four are answered in **file 10**. Do not walk in without them.

2. What the job description is really telling you

I have read a lot of these. Here is what each line means for your interview.

The JD says	What it means for you
"React and Next.js"	Next.js is not optional. They will ask about rendering strategies, and they will notice your Next.js is migration work rather than ownership. File 03 is your biggest opportunity.
"Making Things Fast" (<i>they highlighted this in their own document</i>)	The person who wrote the JD cares about it, and you already have this story (Lighthouse 62 to 88). Files 04 and 10.
"look great on every device"	Responsive CSS, Flexbox and Grid, mobile first. File 05.
"perform well on search engines"	SEO is how this company gets customers. Server rendering, metadata, semantic HTML. Files 03 and 04.
"plug REST APIs and data into the frontend"	fetch and axios, loading and error states, race conditions. File 06.
"reusable components, readable, maintainable code"	They will ask how you would design a component, and probably give you a coding task. Files 07 and 08.
"Partnering with designers and product"	Behavioural round. Have real stories ready. File 09.
18 month commitment, 2 month notice, work from office, 1st and 3rd Saturdays	These will be asked as filter questions. Scripted answers in File 09.

The two things that will set you apart

First, your performance story. Their JD highlights speed. You have a real, measured example of fixing it on a production banking dashboard. Almost nobody at this level has that. Get it into the conversation early.

Second, Next.js reasoning. Almost everyone applying can **use** React. Very few can explain **why Next.js exists** and **what problem server rendering solves for a business that depends on Google search**.

WeMakeScholars finds students through organic search. If your technical answers connect back to *"this helps your pages get found and load fast for students on slow phones"*, you stop sounding like a bootcamp graduate and start sounding like an engineer.

Say it **once**, at the right moment. Not five times.

3. What the interview will probably look like

For a 250 person company hiring at this level, expect 2 or 3 rounds, often on the same day.

Round 1 — Technical screening (45 to 60 min).

JavaScript fundamentals, React hooks, TypeScript, "walk me through your work at Standard Bank", some CSS.

→ Files 01, 02, 05, 10, 11.

Round 2 — Machine coding or a live task (45 to 60 min).

Build a small component. Usually a search and filter list, a form with validation, or fetching and displaying data.

→ File 07.

Round 3 — Tech lead or manager (30 min).

Next.js, performance, SEO, system design, how you work with backend and design.

→ Files 03, 04, 06, 08.

Round 4 — HR (20 min).

Commitment, notice period, salary, working Saturdays.

→ File 09.

4. The schedule — from now until 10:00 AM tomorrow

Time is your scarce resource, so this is ordered by importance. **If you fall behind, drop things from the bottom, never from the top.**

Today, afternoon and evening

Time	What	File
0:00 - 0:15	Read this file	README.md
0:15 - 1:30	React ●	02-react.md
1:30 - 2:45	Next.js ●●	03-nextjs.md

2:45 - 3:30	**Your experience** ●● — the 4 hard questions, out loud	[10-your-projects.md](#)
3:30 - 4:30	**JavaScript** ●	[01-javascript.md](#)
4:30 - 5:15	**TypeScript, state management, testing** ●	[11-typescript-redux-testing.md](#)
5:15 - 6:00	Dinner. Actually eat properly.	—
6:00 - 6:45	**Performance and SEO** ●	[04-performance-seo.md](#)
6:45 - 7:45	**Machine coding** ● — type the code, do not read it	[07-machine-coding.md](#)
7:45 - 8:30	**System design** ●	[08-system-design.md](#)
8:30 - 9:15	**HR answers** — salary, the gap, notice period	[09-hr-and-behavioural.md](#)
9:15 - 9:45	REST APIs, quick read	[06-rest-api-integration.md](#)
By 11:30 PM	**Sleep.** This is not optional.	—

Studying past midnight will make you worse, not better. A tired candidate blanks on things they knew perfectly at 1 AM. Go to bed.

Tomorrow morning

Time	What	File
07:00 - 07:20	Wake up, shower, coffee. No scrolling.	—
07:20 - 08:00	Cheat sheet only	12-final-cheatsheet.md
08:00 - 08:30	HTML and CSS, quick read	05-html-css-responsive.md
08:30 - 09:00	Say your intro and the 4 hard questions out loud, twice	10-your-projects.md
09:00 - 09:20	Re-read the questions you will ask them	09-hr-and-behavioural.md \$6
09:20 - 09:40	Check laptop, charger, internet, ID, resume copies, water	—
09:40 - 09:55	Close everything. Sit up straight. Breathe.	—
10:00	Go.	—

Rule for tomorrow morning: do not learn anything new. New material after 7 AM only pushes out what you already know and makes you anxious. Revision only.

5. The files

File	Topic	Priority
01-javascript.md	Closures, event loop, async, <code>this</code> , ES6, debounce	● High
02-react.md	Hooks, re-renders, keys, context, controlled inputs	● Highest
03-nextjs.md	CSR / SSR / SSG / ISR, App Router, Server Components, SEO	● Highest
04-performance-seo.md	Core Web Vitals, how to diagnose slowness, SEO checklist	● High
05-html-css-responsive.md	Semantic HTML, box model, Flexbox vs Grid, mobile first	● Medium
06-rest-api-integration.md	fetch vs axios, the four states, race conditions, CORS	● Medium
07-machine-coding.md	8 coding problems with full working solutions	● High
08-system-design.md	How to answer design questions, with 8 worked examples	● High
09-hr-and-behavioural.md	Salary, notice period, STAR stories, questions to ask	● High
10-your-projects.md	Your Standard Bank stories + the 4 hard resume questions	●● Highest
11-typescript-redux-testing.md	TypeScript, Redux, Zustand, React Query, testing	● High
12-final-cheatsheet.md	One page. Tomorrow morning only.	● Read last

There are also PDF versions in [pdf/](#). The cheat sheet is designed to be printed.

6. Three rules for the interview room

1. Never just say "I don't know" and stop.

Say this instead:

"I have not used that directly. My understanding is that it does X. Is that the direction you mean?"

Interviewers hire people who can reason out loud. They do not hire people who can only recite.

2. Keep talking during the coding round.

Silence makes them think you are stuck. Narrate what you are doing:

"I will hold the input in state, debounce it, then filter the list..."

3. Connect one answer to their business.

"Since students find you through Google, server rendering matters more here than it would in an internal dashboard."

One sentence like that is what makes an interviewer remember you out of forty CVs.

7. A note on accuracy

File 10 is built from your actual resume, so every story in it should match what the interviewer is reading. If any detail there is wrong, fix it tonight, because they will ask about the exact lines.

Two things to check before the interview:

- **Your QKart repo and live demo both load.** A dead link is worse than no link.
- **You can explain how every number on your resume was measured.** If you cannot defend a number, do not repeat it unprompted.

01 · JavaScript Core

Time needed: 60 minutes

Every frontend interview starts here. React and Next.js are both written in JavaScript. If your basics are weak, the interviewer will stop early and the rest of the interview will not happen.

How to read this file: each question has four parts.

What it means (the idea in plain words) → *Simple example* (small code) →

Why it matters (the real problem it solves) → *Say this* (your answer in the room).

First, the context: what is JavaScript doing on a web page?

A web page has three parts.

Part	Language	Job
Structure	HTML	What is on the page. Headings, buttons, forms.
Look	CSS	Colours, spacing, layout.
Behaviour	JavaScript	What happens when the user does something.

HTML and CSS cannot make decisions. They cannot fetch data. JavaScript can. So when a student clicks "Check loan offers", JavaScript is the part that reads the form, calls the server, waits for the answer, and updates the page.

JavaScript runs **inside the browser**. Every browser has a JavaScript engine built in. This is why it is the only language that runs on a web page directly.

Q1. What is the difference between `var`, `let` and `const`? ●

What it means

These are three ways to create a variable. A variable is a name for a value.

```
var a = 1;    // old way, from 1995
let b = 2;    // new way, when the value will change
const c = 3;  // new way, when the value will not change
```

The difference that matters: scope

Scope means "which part of the code can see this variable".

`let` and `const` live only inside the nearest `{ }` block.

`var` ignores blocks. It leaks out to the whole function.


```
function test() {
  if (true) {
    var x = 'I leak';
    let y = 'I stay inside';
  }
  console.log(x); // "I leak" ← still visible, this is confusing
  console.log(y); // ReferenceError ← y is gone, this is correct
}
```

The classic bug this causes

```
for (var i = 0; i < 3; i++) {
  setTimeout(() => console.log(i), 100);
}
// prints 3, 3, 3 ← WRONG

for (let i = 0; i < 3; i++) {
  setTimeout(() => console.log(i), 100);
}
// prints 0, 1, 2 ← correct
```

With `var` there is only **one** `i` for the whole loop. By the time the timers run, the loop has finished and `i` is 3. With `let`, each turn of the loop gets its own fresh `i`.

Careful: `const` does not mean "cannot change"

`const` only means you cannot point the name at something new. If the value is an object or array, the **inside** can still change.

```
const user = { name: 'Sujith' };
user.name = 'Kumar';           // <span class="tick">✓</span> allowed. We changed what is inside.
user = { name: 'Kumar' };      // ✗ error. We tried to point `user` somewhere new.
```

Say this

"`var` is function scoped, so it leaks out of `if` blocks and loops. `let` and `const` are block scoped, so they stay inside the braces. I use `const` by default, `let` only when I need to reassign, and I do not use `var`. One thing to note is that `const` does not make an object immutable. It only stops you from reassigning the variable itself."

Q2. What is a closure? ●

What it means

A closure is a function that **remembers the variables around it**, even after the outer function has finished running.

Think of it like a lunchbox. When you create the inner function, it packs a copy of the surrounding variables. Wherever it goes later, it still has that lunchbox.

Simple example

```
function makeCounter() {  
  let count = 0;           // this variable lives inside makeCounter  
  
  return function () {  
    count = count + 1;  
    return count;  
  };  
}  
  
const counter = makeCounter();  
counter(); // 1  
counter(); // 2  
counter(); // 3
```

`makeCounter()` already finished on the first line. Normally `count` would be deleted. But the returned function still needs it, so JavaScript keeps it alive. That is a closure.

Notice something useful: nothing outside can touch `count`. You cannot write `counter.count = 100`. The variable is private.

Where you actually use closures

1. **Private data.** As above. The only way to change `count` is through the function.
2. **useState in React.** React stores your state in a closure and hands you back a value and a setter function.
3. **Debounce and throttle.** The timer id is kept in a closure between calls. (See Q10.)
4. **Event handlers that remember which item they belong to.**

The trap they may ask about: stale closure

```
useEffect(() => {  
  const id = setInterval(() => {  
    setCount(count + 1); // ❌ `count` is stuck at its first value, 0  
  }, 1000);  
  return () => clearInterval(id);  
}, []); // empty array = this runs once
```

The function inside `setInterval` packed its lunchbox on the first render only. It will always see `count` as 0. So the counter goes 0 → 1, 1, 1, 1 forever.

Fix: use the function form of the setter. It always receives the newest value.

```
setCount(c => c + 1); // <span class="tick">✓</span>
```

Say this

"A closure is a function that keeps access to the variables from where it was created, even after that outer function has returned. The counter example is the simplest one. In real code I use it for private state and for debounce, where the timer id has to survive between calls. It is also why the stale closure bug happens in `useEffect` with an empty dependency array."

Q3. What is the event loop? ●

The problem it solves

JavaScript can only do **one thing at a time**. It has a single call stack.

So what happens when you call an API that takes 2 seconds? If JavaScript simply waited, the whole page would freeze. No clicking, no scrolling, nothing.

It does not wait. This is what the event loop is for.

How it works, step by step

1. Your code runs on the **call stack**, one line at a time.
2. Slow things (network calls, timers) are **not** run by JavaScript. They are handed to the browser, which handles them in the background.
3. When the browser finishes, it puts your callback function in a **queue**.
4. The **event loop** watches the call stack. The moment the stack is empty, it takes the first item from the queue and runs it.

A simple picture:

```
your code → [ CALL STACK ] ← event loop pushes callbacks back in
                                     ↑
slow work → [ browser handles it ] → [ QUEUE ]
```

The one detail interviewers test

There are two queues, and one has priority.

- **Microtask queue** — Promises, `async/await`. **Higher priority.**
- **Macrotask queue** — `setTimeout`, `setInterval`, click events. Lower priority.

All microtasks run before the next macrotask.

```
console.log('1');
setTimeout(() => console.log('2'), 0);
Promise.resolve().then(() => console.log('3'));
console.log('4');
```

Output: **1, 4, 3, 2**

Why:

- `1` and `4` are normal code, so they run first, in order.
- `3` is a Promise, so it goes to the microtask queue.
- `2` is a `setTimeout`, so it goes to the macrotask queue.
- Microtasks win, so `3` runs before `2`. Even though the timeout was 0 ms.

Say this

"JavaScript is single threaded, so it runs one thing at a time. Slow work like network calls is given to the browser instead of blocking the stack. When it finishes, the callback waits in a queue, and the event loop pushes it back onto the stack once the stack is empty. The detail people miss is that promises go to the microtask queue, which is drained fully before any `setTimeout` runs."

Q4. Callbacks, Promises and async/await •

The context: three generations of the same idea

All three solve one problem. *How do I run some code later, after a slow thing finishes?*

1. Callback (oldest). Pass a function in. It gets called when the work is done.

```

getUser(1, function (user) {
  getOrders(user.id, function (orders) {
    getDetails(orders[0], function (details) {
      console.log(details);      // three levels deep already
    });
  });
});
});

```

This shape is called **callback hell**. It grows sideways and is hard to read.

2. Promise. An object that represents a value you will get later.

It has three states: `pending` → then either `fulfilled` or `rejected`.

```

getUser(1)
  .then(user => getOrders(user.id))
  .then(orders => getDetails(orders[0]))
  .then(details => console.log(details))
  .catch(err => console.log('Something failed:', err));

```

Flat, not nested. One `.catch` handles errors from any step.

3. async/await (newest). The same Promise, written so it reads like normal code.

```

async function show() {
  try {
    const user    = await getUser(1);
    const orders  = await getOrders(user.id);
    const details = await getDetails(orders[0]);
    console.log(details);
  } catch (err) {
    console.log('Something failed:', err);
  }
}

```

`await` means "pause here until this promise finishes". It does not freeze the browser. It only pauses this one function.

Real fetch example, with the mistake people make

```
async function getLoans() {
  const res = await fetch('/api/loans');

  // <span class="warn"><span class="warn">△</span></span> fetch does NOT throw an error on 404 or 500.
  // It only throws if the network itself failed.
  // So you must check this yourself:
  if (!res.ok) {
    throw new Error('Request failed with status ' + res.status);
  }

  return res.json();
}
```

This catches many candidates. Remember it.

Running calls in parallel ●

```
// SLOW: 300ms + 300ms = 600ms
const banks = await getBanks();
const offers = await getOffers();

// FAST: both start together, takes about 300ms
const [banks, offers] = await Promise.all([getBanks(), getOffers()]);
```

Use `Promise.all` when the calls do not depend on each other.

Method	What it does
<code>Promise.all</code>	Waits for all. If any one fails, the whole thing fails.
<code>Promise.allSettled</code>	Waits for all. Tells you which passed and which failed.
<code>Promise.race</code>	Returns the first one to finish, success or failure.
<code>Promise.any</code>	Returns the first one to succeed .

Say this

"Callbacks came first but nesting them becomes unreadable. Promises fixed the nesting with chaining. `async/await` is the same promise written in a way that reads top to bottom, with normal `try/catch` for errors. One thing I always remember is that `fetch` does not reject on a 404 or 500, so I check `res.ok` myself. And when two calls do not depend on each other I use `Promise.all` so they run in parallel instead of one after the other."

Q5. What is `this`? How are arrow functions different? ●

What it means

`this` is a word that points to an object. Which object it points to is decided by **how the function is called**, not where it was written.

The four rules

```
const user = {
  name: 'Sujith',
  greet() { console.log(this.name); }
};

user.greet();           // "Sujith" → called on user, so `this` = user

const g = user.greet;
g();                    // undefined → called alone, so `this` is not user

g.call(user);           // "Sujith" → we forced `this` to be user

new Person();           // `this` = the new object being created
```

Arrow functions are different

A normal function decides `this` when it is called.

An arrow function does **not** have its own `this`. It borrows it from the code around it, at the time it was written. This can never be changed.

```
const user = {
  name: 'Sujith',
  regular: function () { console.log(this.name); },
  arrow: () => { console.log(this.name); }
};

user.regular(); // "Sujith" ← this = user
user.arrow();   // undefined ← this came from outside the object
```

Why this is useful

Before arrow functions, this was a common bug:

```
const timer = {
  seconds: 0,
  start: function () {
    setInterval(function () {
      this.seconds++; // ❌ `this` is not `timer` here
    }, 1000);
  }
};
```

The fix used to be `.bind(this)`. Now we just use an arrow function:

```
setInterval(() => { this.seconds++; }, 1000); // <span class="tick">✓</span> borrows `this` from
start()
```

Say this

"`this` depends on how a function is called. If you call `obj.method()`, `this` is `obj`. If you pull the same function out and call it alone, `this` is lost. Arrow functions do not have their own `this`. They take it from the surrounding code, which is why they solved the old `bind(this)` problem in callbacks. In React with hooks I rarely deal with `this` at all now."

Q6. ES6 features you use every day

These are just shortcuts. Learn what each one replaces.

Destructuring — pull values out of an object or array.

```
// instead of:
const name = user.name;
const city = user.city;

// write:
const { name, city } = user;
const [first, second] = myArray;
```

Default value — used when the value is `undefined`.

```
const { city = 'Hyderabad' } = user;
```

Spread `...` — copy an object or array into a new one.


```
const newUser = { ...user, verified: true }; // copy + change one field
const newList = [...list, newItem]; // copy + add one item
```

⚠ Spread makes a **shallow** copy. Objects inside are still shared.

For a full copy: `structuredClone(user)`.

Rest `...` — collect the remaining items.

```
const [first, ...others] = [1, 2, 3, 4]; // first = 1, others = [2, 3, 4]
```

Template literal — build a string without `+`.

```
const msg = `Hello ${name}, you have ${count} offers`;
```

Optional chaining `?.` — stop safely if something is missing.

```
const city = user?.address?.city; // undefined instead of a crash
```

Nullish coalescing `??` — a fallback, but only for `null` and `undefined`.

```
const rate = apiRate ?? 8.5;
```

● **`??` vs `||` — know this difference**

```
const rate = 0;

rate || 8.5 // 8.5 ← wrong! 0 is falsy, so it was replaced
rate ?? 8.5 // 0 ← correct, 0 is a real value
```

`||` replaces **any** falsy value: `0`, `''`, `false`, `null`, `undefined`, `NaN`.

`??` replaces **only** `null` and `undefined`.

If an interest rate of 0% is valid, `||` will silently give you the wrong number.

Q7. `==` vs `===`, and `null` vs `undefined`

`===` compares value and type. No conversion.

`==` converts the types first, which gives strange results.

```
'5' === 5      // false   ← different types
'5' == 5       // true    ← '5' was converted to 5
0 == ''        // true
null == undefined // true
null === undefined // false
```

Rule: always use `===`.

undefined — JavaScript's default. The variable exists but was never given a value.

null — you deliberately said "empty on purpose".

Falsy values, memorise the list:

`false`, `0`, `''` (empty string), `null`, `undefined`, `NaN`.

Everything else is truthy, including `[]` and `{}`.

Q8. Array methods: map, filter, reduce, forEach •

These come up constantly, because React lists are built with `map`.

```
const loans = [
  { id: 1, bank: 'SBI', amount: 500000 },
  { id: 2, bank: 'HDFC', amount: 800000 },
  { id: 3, bank: 'SBI', amount: 300000 },
];
```

map — change every item. Returns a **new array of the same length**.

```
const names = loans.map(loan => loan.bank); // ['SBI', 'HDFC', 'SBI']
```

This is what you use to turn data into JSX in React.

filter — keep only some items. Returns a **shorter array**.

```
const sbi = loans.filter(loan => loan.bank === 'SBI'); // 2 items
```

This is what you use for search and filter features.

find — get the **first matching item** itself, not an array.

```
const one = loans.find(loan => loan.id === 2); // { id: 2, ... }
```

reduce — turn the whole array into a single value.

```
const total = loans.reduce((sum, loan) => sum + loan.amount, 0);
// 1600000
```

Read it as: start at 0, then add each loan's amount to the running total.

`forEach` — just loop. Returns nothing. Use it only for side effects.

Why this matters in React

`map`, `filter` and `reduce` do **not** change the original array. They return a new one. React needs a new array to notice that something changed. So these methods fit React perfectly, and a loop with `push` does not.

Q9. Event bubbling and event delegation

Bubbling

When you click a button inside a div, the click event fires on the button first, then travels **up** to the div, then to the body. This travelling up is bubbling.

```
<div onClick="...">      ← 3rd
  <ul onClick="...">      ← 2nd
    <li onClick="...">    ← 1st (you clicked here)
```

Delegation, and why it is useful

Imagine a list of 500 loan rows, each with a delete button.
You could add 500 click listeners. That is slow and uses memory.

Instead, add **one** listener on the parent, and check what was clicked:

```
list.addEventListener('click', (e) => {
  const btn = e.target.closest('button');
  if (!btn) return;           // they clicked somewhere else
  deleteRow(btn.dataset.id);
});
```

One listener instead of 500. It also works for rows added later, because the parent was already listening.

Two methods to know

- `e.stopPropagation()` — stop the event from bubbling further up.
- `e.preventDefault()` — stop the browser's default action, like a form reloading the page or a link navigating away.

Q10. Debounce and throttle • (practise writing this)

The problem

A user types "axis bank" in a search box. That is 9 characters. If you call the API on every keystroke, you send **9 requests**. Eight of them are wasted, and the results may arrive out of order.

Debounce: wait until the user stops

"Only run after there has been no typing for 400ms."

Use for: **search boxes**, autocomplete, saving a draft.

```
function debounce(fn, delay = 400) {  
  let timer; // kept alive by a closure  
  
  return function (...args) {  
    clearTimeout(timer); // cancel the previous plan  
    timer = setTimeout(() => fn(...args), delay); // make a new plan  
  };  
}
```

Every keystroke cancels the previous timer. Only the last one survives.

Result: 9 keystrokes, **1 request**.

Throttle: run at most once every X ms

"No matter how many times this fires, run it only once per 300ms."

Use for: **scroll**, resize, mouse move, infinite scroll.

```
function throttle(fn, limit = 300) {  
  let waiting = false;  
  
  return function (...args) {  
    if (waiting) return; // still in the cool-down period  
    fn(...args);  
    waiting = true;  
    setTimeout(() => { waiting = false; }, limit);  
  };  
}
```

The difference in one line

Debounce waits for a pause. Throttle limits the rate.

Say this

"Debounce runs the function only after the user stops for a set time, which is what you want for a search box. Throttle runs it at a fixed maximum rate, which is what you want for scroll events. Without debouncing, a nine letter search sends nine API calls instead of one."

Q11. Hoisting

Before running your code, JavaScript scans it and makes room for all the declarations. This is called hoisting.

```
sayHi(); // <span class="tick">✓</span> works
function sayHi() { ... } // function declarations are fully hoisted

console.log(a); // undefined ← var is hoisted but empty
var a = 5;

console.log(b); // ✗ ReferenceError
let b = 5; // let/const are hoisted but locked
```

That locked period for `let` and `const` is called the **Temporal Dead Zone (TDZ)**. It is a good thing. It gives you a clear error instead of a silent `undefined`.

Q12. Shallow copy vs deep copy •

```
const original = { name: 'Sujith', address: { city: 'Hyderabad' } };

const shallow = { ...original };
shallow.name = 'Kumar'; // <span class="tick">✓</span> original.name is safe
shallow.address.city = 'Chennai'; // ✗ original.address.city ALSO changed

const deep = structuredClone(original);
deep.address.city = 'Chennai'; // <span class="tick">✓</span> original is safe
```

Spread copies only the top level. Anything nested is still the same object in memory. This is the reason behind a very common React bug: "I changed the state but the screen did not update", or "I changed one item and another item changed too".

Q13. localStorage vs sessionStorage vs cookies

	Size	How long it lasts	Sent to the server?
localStorage	about 5 MB	Forever, until cleared	No
sessionStorage	about 5 MB	Until the tab is closed	No
Cookie	about 4 KB	Until it expires	Yes, on every request

Which one for a login token?

An **httpOnly cookie** is the safest. `httpOnly` means JavaScript cannot read it. So if an attacker injects a script into your page, that script still cannot steal the token. Anything in `localStorage` can be read by any script on the page.

For a product handling student loan data, this is worth saying out loud.

⚠ In **Next.js**: `localStorage` does not exist on the server. Reading it in a Server Component or during the first render will crash. Read it inside `useEffect`, which only runs in the browser.

Q14. Quick questions they fire at the end

- **slice vs splice?** `slice` returns a copy and leaves the original alone. `splice` changes the original array.
- **typeof null?** `'object'`. This is a famous bug in JavaScript from 1995. It was never fixed because too much code depends on it.
- **How to check for an array?** `Array.isArray(x)`. `typeof []` gives `'object'`.
- **NaN === NaN?** `false`. NaN is not equal to anything, including itself. Use `Number.isNaN(x)`.
- **Higher order function?** A function that takes a function, or returns one. `map`, `filter` and `debounce` are all higher order functions.
- **Pure function?** Same input always gives the same output, and it changes nothing outside itself. Easy to test and easy to reason about.
- **Synchronous vs asynchronous?** Synchronous blocks the next line until it finishes. Asynchronous starts the work and lets the next line run.

✓ Check yourself before moving on

You should be able to do these **without looking**:

1. Explain a closure using the counter example, in about 30 seconds.
2. Explain why `1 4 3 2` is the answer in the event loop question.
3. Write the `debounce` function from memory.

4. Explain why `0 || 8.5` gives the wrong answer and `0 ?? 8.5` gives the right one.

02 · React (Most important technical file)

Time needed: 75 minutes

This is the main skill for the job. Expect 15 to 20 minutes of React questions, plus a coding task built on React.

| First, the context: what problem does React solve?

Life before React

To change something on a page, you had to find the element and change it by hand:

```
document.getElementById('count').innerText = count;
document.getElementById('warning').style.display = count > 5 ? 'block' : 'none';
document.getElementById('btn').disabled = count === 0;
```

This is called **imperative** code. You give step by step instructions.

The problem is keeping everything in sync. If `count` changes in five places, you must remember to update all three lines in all five places. Miss one and the screen shows the wrong thing. In a big app this becomes impossible to manage.

What React does instead

You describe **what the screen should look like for a given state**. React works out what to change in the page.

```
function Counter({ count }) {
  return (
    <div>
      <p>{count}</p>
      {count > 5 && <p>That is a lot</p>}
      <button disabled={count === 0}>Reset</button>
    </div>
  );
}
```

You never touch the DOM. You change `count`, and React updates the screen. This is called **declarative** code.

The second big idea: components

A component is a reusable piece of UI. Write `<LoanCard />` once, use it on the home page, the search page and the comparison page. This is what the job description means by "reusable components".

Say this

"React is declarative. Instead of writing instructions to change the DOM, I describe what the UI should look like for the current state, and React updates the DOM for me. That removes a whole class of bugs where the screen and the data go out of sync. The component model also means I build something like a loan card once and reuse it everywhere."

Q1. What is the Virtual DOM? ●

What it means

The real DOM is the actual page in the browser. Changing it is **slow**, because the browser may have to recalculate layout and repaint pixels.

The Virtual DOM is a **plain JavaScript copy** of what the page should look like. It is just an object in memory, so making it is cheap and fast.

How an update works

1. State changes.
2. React builds a **new** Virtual DOM tree.
3. React compares it with the **previous** tree. This comparison is called **diffing**.
4. React finds the smallest set of real changes and applies only those.

Example: a list of 100 rows and you change one name. React does not rebuild 100 rows in the browser. It changes one piece of text.

The rules React uses when comparing

- Different element type (`<div>` became `<span>`)? Throw away the old one, build new.
- Same type? Keep the element, only update the attributes that changed.
- For lists, match items by their **key** .

A trick question: "Is the Virtual DOM faster than direct DOM manipulation?"

The honest answer scores better than "yes".

"Not always. Hand written DOM code can be faster if it is done perfectly. The Virtual DOM gives you very good performance without you having to think about it, and it makes the code much easier to maintain. That trade is worth it."

Q2. Why do lists need a `key`? Why is index a bad key? ●

What a key does

A key is a **name tag** for each item in a list. It tells React "this is the same item as before" across renders.

```
{loans.map(loan => <LoanCard key={loan.id} loan={loan} />)}
```

Without keys, React can only go by position. If you insert an item at the top, React thinks every item changed, and does far more work than needed.

Why index is dangerous, with a concrete example

Say you have three text inputs:

```
index 0 → Apple    (user typed "hello" in this box)
index 1 → Banana
index 2 → Cherry
```

Now delete **Apple**. The array becomes `[Banana, Cherry]`.

With `key={index}`:

```
index 0 → Banana   ← React thinks: "item 0 is still here, its text just changed"
index 1 → Cherry
```

React **reuses** the first input box. The text "hello" the user typed is still sitting in it, but now it is next to Banana. The data moved but the DOM did not.

With `key={item.id}`:

React sees that the item with id "apple" is gone. It removes that input box completely, along with the typed text. Correct behaviour.

When is index acceptable?

Only when the list never changes order, never gets items removed, and has no internal state. A static footer menu is fine. Anything the user can edit is not.

Q3. Props vs State ●

The simple difference

- **Props** come from **outside** (the parent). The child cannot change them.
- **State** lives **inside** the component. The component can change it.

Think of a component as a function. Props are the arguments passed in. State is a variable it keeps for itself.

```
function LoanCard({ bank, rate }) { // props, given by the parent
  const [expanded, setExpanded] = useState(false); // state, owned here

  return (
    <div>
      <h3>{bank} – {rate}%</h3>
      <button onClick={() => setExpanded(!expanded)}>
        {expanded ? 'Hide' : 'Show'} details
      </button>
    </div>
  );
}
```

How does a child send data back to the parent?

Data flows **down** through props. Events flow **up** through callbacks.
The parent passes a function down, and the child calls it.

```
// Parent
<SearchBar onSearch={(text) => setQuery(text)} />

// Child
<input onChange={(e) => props.onSearch(e.target.value)} />
```

"Lifting state up"

If two sibling components need the same value, neither can own it. Move the state **up** to their closest shared parent, then pass it down to both.

```

      Parent ← state lives here
    /      \
  SearchBar ResultList

```

Q4. `useState` — the three traps ●

```
const [count, setCount] = useState(0);
//   ↑ value   ↑ function to change it   ↑ starting value
```

Trap 1: state updates are not instant

```
setCount(count + 1);
setCount(count + 1);
// count only goes up by 1, not 2
```

Both lines read the same old `count`. React batches the updates and applies them after the current work finishes.

Fix — use the function form. React passes you the latest value:

```
setCount(c => c + 1);  
setCount(c => c + 1);  
// now it goes up by 2 <span class="tick">✓</span>
```

Rule: if the new value depends on the old value, use the function form.

Trap 2: never change state directly

```
items.push(newItem);  
setItems(items);           // ✗ screen does not update
```

React checks "is this the same object as before?" It is the same array, just with one more item inside. So React decides nothing changed and skips the re-render.

```
setItems([...items, newItem]);           // <span class="tick">✓</span> a brand new array  
setUser({ ...user, verified: true });    // <span class="tick">✓</span> a brand new object
```

Always create a new array or object.

Trap 3: expensive starting values

```
useState(readFromLocalStorage())         // ✗ runs on every single render  
useState(() => readFromLocalStorage())    // <span class="tick">✓</span> runs once, on the first render
```

Q5. `useEffect` ●

What it is for

`useEffect` is for talking to things **outside** React. For example:

- fetching data from an API
- setting a timer
- adding a browser event listener
- saving to localStorage

The dependency array controls when it runs

```
useEffect(() => { ... });           // after EVERY render (rarely what you want)
useEffect(() => { ... }, []);        // once, when the component appears
useEffect(() => { ... }, [userId]); // when userId changes
```

Cleanup: the part people forget

The function you `return` from an effect is the cleanup. React runs it when the component disappears, and before running the effect again.

```
useEffect(() => {
  const id = setInterval(tick, 1000);

  return () => clearInterval(id); // ← cleanup
}, []);
```

Without cleanup, the timer keeps running after the component is gone. It tries to update a component that no longer exists. That is a memory leak, and you will see warnings in the console.

Things that need cleanup: timers, event listeners, API subscriptions, and in flight fetch requests.

● When should you NOT use useEffect?

This question separates good candidates from average ones.

Do not use an effect for data you can calculate.

```
// ❌ Wrong: extra state, extra render, and it can go out of sync
const [filtered, setFiltered] = useState([]);
useEffect(() => {
  setFiltered(items.filter(i => i.bank === bank));
}, [items, bank]);

// <span class="tick">✓</span> Right: just calculate it while rendering
const filtered = items.filter(i => i.bank === bank);
```

The second version is shorter, has no extra render, and can never be stale.

Why does my effect run twice in development?

React 18 StrictMode mounts your component, unmounts it, and mounts it again on purpose. It does this to reveal missing cleanup. It only happens in development, never in production.

Q6. The hooks list

Hook	What it is for
<code>useState</code>	Store a value that changes and should re-render the screen
<code>useEffect</code>	Talk to the outside world (API, timers, browser)
<code>useContext</code>	Read shared data without passing props down every level
<code>useRef</code>	Store a value that survives renders but does not re-render
<code>useMemo</code>	Remember a calculated value so it is not recalculated
<code>useCallback</code>	Remember a function so it is not recreated
<code>useReducer</code>	Manage complex state with many different actions
Custom hook	Your own reusable <code>useSomething</code> function

• The Rules of Hooks, and the reason behind them

Only call hooks at the top level. Never inside an `if`, a loop, or a nested function.

```
if (isLoggedIn) {  
  const [name, setName] = useState(''); // ❌ never do this  
}
```

Why? React does not know your hooks by name. It tracks them by **the order they are called**. First hook, second hook, third hook. If an `if` skips one, the order shifts, and React gives you the wrong value from the wrong hook.

Q7. `useMemo`, `useCallback` and `useRef` •

`useMemo` — remember a value

```
const total = useMemo(() => {  
  return loans.reduce((sum, l) => sum + l.amount, 0);  
}, [loans]);
```

The calculation only runs again when `loans` changes. On other renders React reuses the stored answer.

useCallback — remember a function

```
const handleAdd = useCallback((id) => {  
  setCart(c => [...c, id]);  
}, []);
```

Why would a function need remembering? Every render creates a **new** function object. Even though the code is identical, it is a different object in memory. If you pass it to a child wrapped in `React.memo`, the child sees a "new" prop and re-renders anyway. `useCallback` keeps the same function object.

useRef — a box that does not trigger re-renders

Two uses.

1. Reach a DOM element:

```
const inputRef = useRef(null);  
useEffect(() => { inputRef.current.focus(); }, []);  
return <input ref={inputRef} />;
```

2. Keep a value between renders without re-rendering:

```
const timerId = useRef(null); // storing a timer id  
const isFirstRender = useRef(true); // a flag
```

useState vs useRef in one line: changing state re-renders the screen. Changing a ref does not.

● "Should you wrap everything in useMemo?"

Say **no**. This is a maturity signal.

"No. Memoization is not free. React still has to store the value and compare the dependencies on every render. If the calculation is cheap, memoizing it can be slower than just doing it. I profile with React DevTools first and memoize where there is a real problem."

Q8. What causes a re-render? How do you stop unnecessary ones? ●

A component re-renders when:

1. Its own state changed.
2. Its props changed.
3. **Its parent re-rendered.** This happens even if its props are identical.

4. A context it uses changed.

Point 3 surprises people. By default, when a parent re-renders, all its children re-render too.

How to fix it, in the order you should try

1. `React.memo` — skip the re-render if the props are the same.

```
const LoanCard = React.memo(function LoanCard({ loan }) { ... });
```

2. `useCallback` / `useMemo` on the props you pass to it.

`React.memo` compares props. If you pass a fresh function or object every render, the comparison always fails and `React.memo` does nothing.

3. **Move state down.** If only a small part of the page uses a piece of state, put that state in a smaller component. Then only that part re-renders.

4. **Split your contexts.** If one context holds both the theme and the cart, every cart change re-renders everything using the theme too.

Q9. Controlled vs uncontrolled inputs •

Controlled — React state holds the value. React is the source of truth.

```
const [email, setEmail] = useState('');

```

Uncontrolled — the DOM holds the value. You read it when you need it.

```
const emailRef = useRef();

```

Which one, and why

Use **controlled** by default. Because the value is in state, you can:

- validate as the user types
- disable the submit button when the form is invalid
- format the value, for example adding commas to an amount
- reset the form easily

Use **uncontrolled** for very large forms where a re-render on every keystroke is too slow, or for file inputs. `<input type="file">` is always uncontrolled, because for security reasons JavaScript cannot set its value.

Q10. Context API •

The problem it solves: prop drilling

The logged in user is needed deep in the tree. Without context you pass it through every level, even components that do not use it:

```
App (has user)
├─ Layout      (passes user, does not use it)
│   └─ Sidebar (passes user, does not use it)
│       └─ Menu (passes user, does not use it)
│           └─ Avatar ← finally uses it
```

How context fixes it

```
const AuthContext = createContext(null);

// near the top
<AuthContext.Provider value={{ user, logout }}>
  <Layout />
</AuthContext.Provider>

// anywhere below, at any depth
const { user } = useContext(AuthContext);
```

Good for values that rarely change: the current user, the theme, the language.

• The limitation you must mention

Every component using a context re-renders whenever the context value changes.

And this is a common mistake:

```
<AuthContext.Provider value={{ user, logout }}>
```

That object literal is **created fresh on every render**. So it is always "new", and every consumer re-renders every time, even if `user` did not change.

Fix:

```
const value = useMemo(() => ({ user, logout }), [user]);
<AuthContext.Provider value={value}>
```

"Context solves prop drilling, but it is not a state manager. It is not optimised for values that change often, because every consumer re-renders. For server data I would use React Query instead, and for large client state, Redux Toolkit or Zustand."

Q11. Custom hooks

A custom hook is just a function whose name starts with `use` and which calls other hooks. It lets you reuse **logic**, not UI.

```
function useDebounce(value, delay = 500) {
  const [debounced, setDebounced] = useState(value);

  useEffect(() => {
    const timer = setTimeout(() => setDebounced(value), delay);
    return () => clearTimeout(timer);
  }, [value, delay]);

  return debounced;
}

// using it
const query = useDebounce(searchText, 400);
useEffect(() => { callApi(query); }, [query]);
```

Important detail: each component that calls `useDebounce` gets its **own separate state**. The hook shares the logic, not the data.

Have one custom hook ready to describe. It is a strong signal.

Q12. Code splitting and error boundaries

Code splitting with lazy and Suspense

By default, all your code is bundled into one big JavaScript file. The user downloads the admin dashboard even if they never open it.

```
const Dashboard = React.lazy(() => import('./Dashboard'));

<Suspense fallback={<Spinner />}>
  <Dashboard />
</Suspense>
```

Now the dashboard code downloads only when it is needed. The initial page loads faster. This connects directly to file 04 on performance.

Error boundary

If any component throws an error during render, React unmounts the whole app and the user sees a blank white page. An error boundary catches that and shows a fallback instead.

```
class ErrorBoundary extends React.Component {
  state = { hasError: false };
  static getDerivedStateFromError() { return { hasError: true }; }
  render() {
    if (this.state.hasError) return <p>Something went wrong.</p>;
    return this.props.children;
  }
}
```

This is the one place where you still write a class component. Hooks cannot do it yet. In practice most teams use the `react-error-boundary` package.

Q13. Class components vs function components

Class (old)	Function + hooks (current)
<code>componentDidMount</code>	<code>useEffect(fn, [])</code>
<code>componentDidUpdate</code>	<code>useEffect(fn, [dep])</code>
<code>componentWillUnmount</code>	the cleanup <code>return</code> inside <code>useEffect</code>
<code>this.state</code> , <code>this.setState</code>	<code>useState</code>

"Function components with hooks are the standard now. There is less boilerplate, no confusion with `this` , and logic can be reused through custom hooks instead of higher order components. I can read class components in older code, but I write function components."

Q14. Which state management would you choose?

Answer with a decision path, not a favourite library:

1. **Local** `useState` first. Most state belongs to one component.
2. **Lift it up** if two siblings need it.

3. **Context** for values that are shared widely and change rarely, like the user or the theme.
4. **React Query or SWR** for server data. Most "global state" is really a copy of server data, and these libraries handle caching, refetching, loading and error states for you.
5. **Redux Toolkit or Zustand** only when client state is genuinely complex.

"I would not add Redux to a project just to hold a user object. That is extra setup with no benefit."

Q15. Quick questions

- **What is JSX?** It looks like HTML but it is JavaScript. A build tool converts `<div>Hi</div>` into `React.createElement('div', null, 'Hi')`. The browser never sees JSX.
- **Why `className` and not `class`?** `class` is a reserved word in JavaScript.
- **What is a Fragment?** `<>...</>` groups elements without adding an extra `<div>` to the page.
- **How do you render conditionally?** `{isLoading && <Spinner />}` or a ternary.
- ⚠ **The `0` trap:** `{count && <p>...</p>}` renders a literal `0` on the screen when count is 0, because `0` is falsy but React still prints it. Write `{count > 0 && <p>...</p>}`.
- **React.memo vs useMemo?** `React.memo` wraps a component. `useMemo` caches a value inside a component.
- **Can you read `key` as a prop?** No. React uses it internally, the component never receives it.

✓ Check yourself before moving on

1. Explain the four things that cause a re-render, and three ways to reduce them.
2. Write `useDebounce` from memory.
3. Explain the "index as key" problem using the delete example, in 30 seconds.
4. Explain why `items.push(x); setItems(items)` does not update the screen.

03 · Next.js (Your best chance to stand out)

Time needed: 75 minutes

The job description names Next.js. Their business depends on Google search. Most candidates at this level can use React but cannot explain **why Next.js exists**. If you can, you move to the top of the list.

| First, the context: how a normal React app loads

When you build a plain React app (Create React App or Vite), the server sends the browser an almost empty file:

```
<html>
  <body>
    <div id="root"></div>           ← completely empty
    <script src="/bundle.js"></script> ← 300 KB of JavaScript
  </body>
</html>
```

Then, in the browser:

1. Download the HTML. It has no content.
2. Download the JavaScript bundle.
3. Parse and run the JavaScript.
4. React builds the page.
5. React asks the API for data.
6. Finally the user sees the loan offers.

This is called **Client Side Rendering (CSR)**. The client (browser) does the work.

Two problems with this

Problem 1: the user waits. On a mid range Android phone on 4G, steps 2 to 6 can take several seconds. The user stares at a blank white screen.

Problem 2: Google sees an empty page. When Google's crawler first visits, the HTML has no content in it. Google can run JavaScript, but it does so later, in a second pass. That makes indexing slower and less reliable. Other crawlers, like LinkedIn and WhatsApp link previews, are much worse at it.

For WeMakeScholars this second problem is serious. Students find them by searching Google for education loans. If the pages do not rank, there are no students.

Q1. Why Next.js instead of plain React? ●●

What Next.js changes

Next.js runs React **on the server first**. The server builds the finished HTML and sends that to the browser.

```
<html>
  <body>
    <h1>Education Loans for Study Abroad</h1> ← real content, immediately
    <ul><li>SBI – 9.15%</li><li>HDFC – 9.55%</li></ul>
    <script src="/bundle.js"></script>
  </body>
</html>
```

The user sees content straight away. Google sees content straight away.

What else you get for free

- **File based routing.** Create a file, get a URL. No router setup.
- **Automatic code splitting.** Each page loads only its own JavaScript.
- `next/image`. Automatic image optimisation.
- **Built in SEO metadata.** Titles and descriptions per page.
- **API routes.** Small backend endpoints inside the same project.

Say this

"A plain React app sends an empty HTML file and then builds the page in the browser. That means the user waits with a blank screen, and a crawler's first look at the page has no content in it. Next.js renders on the server, so the browser gets real HTML immediately. That helps both first paint speed and SEO. It also gives me file based routing, code splitting and image optimisation without extra setup. For a site like WeMakeScholars, where students arrive from Google, that is not optional, it is how the pages get found."

If they push back: "But Google runs JavaScript now"

"It does, but rendering happens in a second pass that is queued, so indexing is slower and less reliable. Other crawlers such as Bing and social preview bots are worse at it. Server rendered HTML removes the risk. And it still helps real users on slow phones, which is the bigger benefit anyway."

Q2. CSR, SSR, SSG, ISR ●● (the most valuable answer in this guide)

These are four different **times** at which the HTML can be built.

1. CSR — Client Side Rendering

Built in the browser, after JavaScript loads.

The plain React behaviour described above.

Good for: pages behind a login, where SEO does not matter. A student's dashboard.

2. SSR — Server Side Rendering

Built on the server, fresh for every single request.

A student opens the page → the server runs the code → sends finished HTML.

Good for: pages that are different for each user, or must be perfectly up to date.

Cost: the server does work on every request, so it is the slowest of the server options and puts the most load on your infrastructure.

```
// App Router
const res = await fetch(url, { cache: 'no-store' }); // no cache = SSR
```

3. SSG — Static Site Generation

Built once, at build time. When you run `npm run build`, Next.js creates the HTML files. They sit on a CDN and are served instantly.

Good for: content that is the same for everyone and rarely changes. An "About us" page, a blog post, a landing page.

Cost: to change the content you must rebuild and redeploy the whole site.

```
const res = await fetch(url); // cached by default = SSG
```

4. ISR — Incremental Static Regeneration

The best of both. Built like SSG, but Next.js rebuilds the page quietly in the background every N seconds.

```
const res = await fetch(url, { next: { revalidate: 3600 } }); // every hour
```

How it works:

- A visitor gets the stored HTML instantly, like SSG.
- If the page is older than one hour, Next.js rebuilds it in the background.
- The next visitor gets the fresh version.
- Nobody waits for the rebuild.

Good for: content that is the same for everyone but changes now and then.

Summary table

	When HTML is made	Speed for user	Fresh data?	Use for
CSR	In the browser	Slowest first paint	Yes	Logged in dashboards
SSR	On every request	Good	Always fresh	Personalised pages
SSG	Once, at build	Fastest	Only at build time	Blog, about page
ISR	At build + refreshed on a timer	Fastest	Fresh within N seconds	Listings, prices

● The follow up question: "Which would you use for their bank listing page?"

This is where you win the interview. Think out loud:

"The list of partner banks and their interest rates is the **same for every visitor**, so SSR would be wasteful. The server would rebuild an identical page for every student who lands on it.

But the rates **do change** sometimes, and rebuilding the whole site for one rate change is not practical.

So I would use **ISR with a revalidate of about an hour**. Visitors get a static file served from the CDN, which is the fastest possible, and the content is never more than an hour old.

The student's own application status page is different. That is personal to them and must be current, so that would be SSR or client side behind a login."

Practise saying that out loud tonight. It is your single strongest answer.

Q3. App Router vs Pages Router ●

Next.js has two routing systems. Version 13 introduced the new one.

	Pages Router (older)	App Router (newer)
Folder	<code>/pages</code>	<code>/app</code>
A page	<code>pages/about.js</code>	<code>app/about/page.js</code>
Get data	<code>getServerSideProps</code> , <code>getStaticProps</code>	<code>async</code> component with <code>await fetch</code>
Default component type	Client	Server Component
Shared layout	<code>_app.js</code>	<code>layout.js</code> , and it can be nested
Loading state	You build it	<code>loading.js</code> file
Error state	<code>_error.js</code>	<code>error.js</code> per folder

Be honest about which one you have used. Then show you understand the other.

"I have built mostly with [X]. I know the App Router is the direction Next.js is going, and the main differences are Server Components by default and data fetching directly in the component instead of `getServerSideProps`."

Q4. Server Components vs Client Components ●●

This is the newest concept, and the one people find confusing. Here is the simple version.

The idea

In the App Router, **components run on the server by default**. Their JavaScript is never sent to the browser at all. Only the HTML they produced is sent.

If a component needs to be interactive (a button, an input, anything with `useState` or `onClick`), you mark it with `'use client'` at the top of the file. That component's code is then sent to the browser.

Example

```
// app/loans/page.js - Server Component (no 'use client')
export default async function LoansPage() {
  // You can await directly. This runs on the server.
  const res = await fetch('https://api.example.com/loans', {
    next: { revalidate: 60 }
  });
  const loans = await res.json();

  return (
    <div>
      <h1>Education Loans</h1>
      <SearchBar />           {/* interactive, so it is a client component */}
      <LoanList loans={loans} /> {/* just displays data, stays on server */}
    </div>
  );
}
```

```
// components/SearchBar.jsx
'use client'; // ← this file goes to the browser

import { useState } from 'react';

export default function SearchBar() {
  const [query, setQuery] = useState('');
  return <input value={query} onChange={e => setQuery(e.target.value)} />;
}
```

What each can do

Server Component	Client Component
Sends zero JavaScript to the browser	Adds to the JavaScript bundle
Can <code>await</code> data directly	Cannot <code>await</code> at the top level
Can read secrets and hit a database safely	Runs in the browser, so no secrets
✗ No <code>useState</code> , <code>useEffect</code> , <code>onClick</code>	✓ All hooks and events work

● The rule to state

"Keep components on the server by default, and push `'use client'` down to the leaves, meaning the small interactive pieces. If I put `'use client'` at the top of the page, every component inside it also goes to the browser and I have thrown away the benefit."

Environment variables (a good detail to know)

In Next.js, environment variables are **server only** by default. That is safe. Only variables starting with `NEXT_PUBLIC_` are sent to the browser.

```
DATABASE_PASSWORD=secret      ← safe, server only
NEXT_PUBLIC_API_URL=https://... ← visible to anyone in the browser
```

Never put a secret in a `NEXT_PUBLIC_` variable. Anyone can read it in DevTools.

| Q5. Routing

The folder structure **is** the routing.

```

app/
  layout.js           → wraps everything (nav, footer)
  page.js             → /
  about/page.js       → /about
  loans/page.js       → /loans
  loans/[id]/page.js  → /loans/123      (dynamic)
  blog/[...slug]/page.js → /blog/a/b/c    (catch all)
  loading.js          → shown while the page loads
  error.js            → shown if the page throws
  not-found.js        → the 404 page
  api/leads/route.js  → an API endpoint at /api/leads

```

Navigation

```

import Link from 'next/link';
<Link href="/loans">View loans</Link>

```

● Why `<Link>` and not `<a>` ?

- `<a href="/loans">` makes the browser throw the whole page away and download everything again from scratch.
- `<Link>` fetches only the new page's data and swaps it in. Much faster.
- `<Link>` also **prefetches** the page when the link scrolls into view, so the navigation feels instant.
- Importantly, `<Link>` still renders a real `<a>` tag underneath. So Google can still follow it and middle click still works.

Programmatic navigation

```

'use client';
import { useRouter } from 'next/navigation'; // App Router

const router = useRouter();
router.push('/dashboard');

```

⚠ App Router uses `next/navigation`. Pages Router uses `next/router`. Mixing them up is a common mistake, so mention that you know the difference.

Q6. `next/image` •

```
import Image from 'next/image';

<Image
  src="/bank-logo.png"
  alt="SBI bank logo"
  width={400}
  height={300}
  priority           // only for the main image at the top of the page
/>
```

What it does automatically

1. **Converts the format.** Serves WebP or AVIF to browsers that support them. These are much smaller than JPEG or PNG.
2. **Resizes.** Generates several sizes and serves the right one for the device. A phone does not download a 2000 pixel wide image.
3. **Lazy loads.** Images below the fold only download when the user scrolls near them.
4. **Reserves space.** Because you gave width and height, the browser leaves an empty box of the right size. The page does not jump when the image arrives. This directly improves your CLS score (see file 04).

`priority` turns **off** lazy loading for your main hero image, so it loads immediately instead of waiting.

The same idea for fonts

`next/font` downloads Google Fonts at build time and serves them from your own server. No extra connection to Google, and no flash of invisible text.

Q7. SEO in Next.js • (their highlighted requirement)

```
// A fixed title, in app/layout.js or any page.js
export const metadata = {
  title: 'Education Loans for Study Abroad | WeMakeScholars',
  description: 'Compare education loan offers from 15+ banks and NBFCs. Free.',
  openGraph: { title: '...', images: ['/og-image.png'] },
  alternates: { canonical: 'https://example.com/loans' },
};
```

```
// A title built from data, for a dynamic page
export async function generateMetadata({ params }) {
  const bank = await getBank(params.id);
  return {
    title: `${bank.name} Education Loan Interest Rates 2026`,
    description: `Current ${bank.name} education loan rates, eligibility and documents.`,
  };
}
```

Also worth naming:

- `app/sitemap.js` and `app/robots.js` — Next.js generates these files for you.
- **JSON-LD structured data** — extra machine readable information in a `<script type="application/ld+json">` tag. It can give you rich results in Google, like an FAQ dropdown under your link. A loan FAQ page is a perfect fit for this.

Q8. What is hydration?

The idea

The server sends finished HTML, so the user sees the page immediately. But that HTML is static. Buttons do not work yet.

Then the JavaScript arrives and React attaches all the event handlers to the existing HTML. That process is called **hydration**. Like adding water to something dried.

There is a short window where the page **looks** ready but is not yet interactive.

Hydration errors

If the HTML the server made does not match what React builds in the browser, you get a hydration mismatch warning.

Common causes:

```
<p>{new Date().toLocaleTimeString()}</p> // server time ≠ browser time
<p>{Math.random()}</p> // different every time
<p>{window.innerWidth}</p> // window does not exist on the server
```

Fix: move it into `useEffect`, which only runs in the browser, or load the component with `next/dynamic` and `{ ssr: false }`.

Q9. API Routes

You can put small backend endpoints inside your Next.js project.

```
// app/api/leads/route.js
export async function POST(request) {
  const body = await request.json();

  // The API key stays on the server. The browser never sees it.
  await fetch('https://crm.internal/leads', {
    method: 'POST',
    headers: { Authorization: `Bearer ${process.env.CRM_SECRET}` },
    body: JSON.stringify(body),
  });

  return Response.json({ ok: true }, { status: 201 });
}
```

Useful when you need to hide an API key, reshape a response, or handle a form submission without building a separate backend service.

Q10. Quick questions

- **next/dynamic ?** Load a component only when needed. With `{ ssr: false }` it skips the server entirely, which you need for libraries that use `window`, like charts and maps.
- **Middleware?** Code that runs before a request is completed. Used for auth redirects and A/B tests.
- **Streaming?** With `loading.js` or `Suspense`, Next.js sends the page shell first and streams in the slow parts as they become ready. The user sees something sooner.
- **How do you check what rendering a page uses?** Run `next build`. It prints a symbol next to each route showing static, SSG or dynamic.
- **Deployment?** Vercel is the simplest since they build Next.js. It also runs on any Node server with `next build && next start`.

✓ Check yourself before moving on

1. Say the CSR / SSR / SSG / ISR table out loud, then pick one for a bank listing page **and give the reason**.
2. Explain Server vs Client Components, and the "push `'use client'` to the leaves" rule.
3. Explain hydration in three sentences.

If your Next.js experience is only from tutorials, say so. Then show the understanding above. Interviewers forgive thin experience. They do not forgive bluffing that falls apart under one follow up question.

04 · Performance and SEO

Time needed: 60 minutes

In their job description, the line "**Making Things Fast**" was highlighted in their own document. That means the person who wrote it cares about it. Treat this as a named requirement, not general knowledge.

| First, the context: why speed is a business problem here

WeMakeScholars gets students from Google search. So two things follow.

1. Speed affects ranking. Google uses page speed as a ranking signal. A slow page appears lower in results, so fewer students find it.

2. Speed affects conversion. Most students are on mid range Android phones on 4G. If the page takes 5 seconds, many leave before it loads. The loan form never gets filled.

So on this product, performance work is **lead generation**, not polish. Say that once in the interview. It reframes you from someone who writes code to someone who understands why the code matters.

| Q1. Core Web Vitals ●●

Google measures three things. Learn all three with their numbers.

LCP — Largest Contentful Paint

"When does the main content appear?"

It measures the time until the biggest thing on screen is visible. Usually the hero image or the main heading.

Good: under 2.5 seconds.

Common causes of a bad score:

- A huge unoptimised hero image
- Slow server response
- CSS or JavaScript blocking the page from rendering

Fixes: optimise and preload the hero image, use SSR or SSG so HTML arrives ready, use a CDN, compress files.

CLS — Cumulative Layout Shift

"Does the page jump around while loading?"

You have felt this. You go to tap a button, an image finishes loading above it, everything shifts down, and you tap the wrong thing.

Good: under 0.1.

Common causes:

- Images without width and height, so no space is reserved
- Banners or ads inserted at the top after load
- A custom font loading late and changing the text size

Fixes: always set width and height on images (or use `next/image`), reserve space for anything that loads late, use `font-display: swap`.

INP — Interaction to Next Paint

"When I tap something, how fast does the page respond?"

This replaced FID in 2024.

Good: under 200 milliseconds.

Cause: JavaScript is busy doing something long, so the browser cannot respond to the tap. Remember from file 01 that JavaScript is single threaded.

Fixes: break long tasks into smaller ones, debounce expensive handlers, reduce unnecessary re-renders, ship less JavaScript.

The one line summary

LCP = is it there. CLS = does it stay still. INP = does it respond.

Two more terms you may hear: **TTFB** (Time to First Byte, how fast your server replies, good is under 800ms) and **FCP** (First Contentful Paint, the first pixel of any content).

| Q2. "The page is slow. How do you find out why?" •

Never answer with a list of fixes. They are testing your **process**.

Step 1: Measure

- **Lighthouse** in Chrome DevTools for a score and a list of problems.
- **Network tab** to see what is being downloaded and how big it is.
- **Performance tab** to record and find long JavaScript tasks.
- Throttle to **Slow 4G** and 4x CPU slowdown, so you see what a real phone sees.

Say the line: *"I do not optimise what I have not measured."*

Step 2: Decide which category the problem is in

Category	Symptom in DevTools
Network	Very large files, or too many requests
Server	Long wait before the first byte arrives (TTFB)
Rendering	CSS or JS blocking, blank screen for a long time
JavaScript	Long tasks in the Performance tab, page feels frozen

Step 3: Fix the biggest item first

Usually images, then bundle size. Do not start with micro optimisations.

Step 4: Measure again

Prove the change worked. Compare the before and after Lighthouse numbers.

Q3. The optimisation toolbox

Learn these in **groups**. Reciting a random list sounds memorised. Grouping sounds like understanding.

Group 1: Images (usually the biggest win)

- **Modern formats.** WebP and AVIF are much smaller than JPEG. `next/image` does this automatically.
- **Right size.** Do not send a 2000 pixel image into a 400 pixel space.
- **Lazy load** anything below the fold, so it only downloads when scrolled to.
- **Set width and height** so the layout does not jump. This fixes CLS.

Group 2: JavaScript

- **Code splitting.** Split by route with `next/dynamic` or `React.lazy`, so the user only downloads the page they are on.
- **Tree shaking.** Import only what you use. `import _ from 'lodash'` pulls in the whole library. `import debounce from 'lodash/debounce'` pulls in one function.
- **Check your dependencies.** Is that date library 70 KB when you only format one date? `date-fns` is smaller than `moment`.
- **Delay third party scripts.** Analytics and chat widgets are often the slowest things on a page. Load them last with `next/script` and `strategy="lazyOnload"`.

Group 3: Rendering

- Use **SSG or ISR** so the HTML is ready before JavaScript loads.

- Reduce re-renders with `React.memo` and `useMemo`, but only where you measured a problem.
- **Virtualise long lists.** With `react-window`, a 5,000 row table only puts about 20 rows in the DOM at a time. The rest are not rendered until you scroll.
- **Debounce** search inputs and **throttle** scroll handlers.

Group 4: Network and delivery

- Use a **CDN** so files are served from a server near the user.
- Enable **gzip or Brotli** compression.
- **Self host fonts** with `next/font` so there is no extra connection.
- **Avoid waterfalls.** Two API calls that do not depend on each other should run with `Promise.all`, not one after the other.

| Q4. Technical SEO for a React or Next app •

The core issue

If content only appears after JavaScript runs, crawlers may not see it reliably. So anything that must rank should be **server rendered**.

The checklist

- **Unique title and meta description on every page.** Use `generateMetadata` in Next.
- **Semantic HTML.** One `<h1>` per page, then `<h2>` and `<h3>` in order. Use `<nav>`, `<main>`, `<article>`. Crawlers use this structure to understand the page.
- **alt text on every meaningful image.** Helps accessibility and image search.
- **Canonical URL** so Google knows which version of a page is the real one, when the same content is reachable from two URLs.
- **sitemap.xml and robots.txt**. Next.js can generate both from files.
- **Structured data (JSON-LD).** Machine readable information about the page. An FAQ page can get an expandable FAQ shown directly in Google results. A loan FAQ is a perfect example, and worth volunteering in the interview.
- **Open Graph tags** so shared links show a proper preview card.
- **Real `<a>` links for internal navigation.** A `<div onClick={router.push}>` is invisible to a crawler. `<Link>` renders a real anchor, so it is fine.
- **Mobile first and fast.** Google indexes the mobile version of your site.

| Q5. Accessibility, briefly

Worth a minute. It overlaps with SEO because both rely on good structure.

- Use real elements. A `<button>` is focusable, works with the Enter key, and is announced correctly by screen readers. A `<div onClick>` does none of that.
- Connect labels to inputs with `htmlFor` and `id`.
- Keep visible focus outlines. Do not remove them with `outline: none`.
- Text contrast of at least 4.5 to 1.
- Use `aria-label` only when there is no visible text, such as an icon only button. ARIA is a patch. Correct HTML is better.
- Quick test: press Tab through the page. Can you reach and use everything?

Q6. Caching, in simple terms

Type	Where	Example
Browser cache	User's device	JS and CSS files, cached for a year
CDN cache	Edge servers worldwide	Static pages served near the user
Data cache	Your app	Next.js <code>revalidate</code> , React Query
Memoization	Inside React	<code>useMemo</code> , <code>React.memo</code>

Useful detail: build tools add a hash to filenames like `main.a3f9c2.js`. So you can cache that file forever. When the code changes, the filename changes, and the browser downloads the new one automatically.

Q7. Quick questions

- **How do you reduce the initial bundle size?** Split code by route, lazy load below the fold components, remove heavy libraries, and check with `@next/bundle-analyzer`.
- **What is lazy loading?** Delaying the download of something until it is actually needed.
- **What is a CDN?** Servers spread around the world that hold copies of your files, so users download from a nearby one instead of a distant origin server.
- **defer vs async on a script tag?** Both download in the background. `async` runs as soon as it arrives, in no guaranteed order. `defer` runs after the HTML is parsed, in order. Use `defer` for anything that touches the page.
- **Reflow vs repaint?** Reflow means the browser recalculates positions and sizes, which is expensive. Repaint means it redraws colours, which is cheaper. This is why you animate `transform` and `opacity` and never `width` or `top`. Those two skip layout entirely and run on the GPU.

✓ Check yourself before moving on

1. Name LCP, CLS and INP with their target numbers and one fix each.
2. Give the four step process for "the page is slow".
3. Explain in one sentence why client side rendering is a risk for SEO.

05 · HTML, CSS and Responsive Design

Time needed: 30 minutes tonight, quick review tomorrow

The first line of the job description is about "translating UI/UX wireframes into clean, performant, responsive code". This topic shows up more in the coding round than in questions, but a few questions are still likely.

Q1. What is semantic HTML and why does it matter? ●

Semantic means the tag describes what the content **is**, not how it looks.

```
<!-- Not semantic: everything is a div -->
<div class="header">
  <div class="nav">
    <div class="btn" onclick="submit()">Apply</div>
  </div>
</div>

<!-- Semantic -->
<header>
  <nav>
    <button onclick="submit()">Apply</button>
  </nav>
</header>
```

Three reasons it matters

1. **Accessibility.** A screen reader user can jump straight to `<nav>` or `<main>`. A `<button>` can be reached with the Tab key and pressed with Enter. A `<div onClick>` cannot do either, unless you add a lot of extra code.
2. **SEO.** Crawlers use the structure to understand what the page is about.
3. **Readability.** The next developer can see the shape of the page instantly.

Tags worth knowing: `<header>`, `<nav>`, `<main>`, `<section>`, `<article>`, `<aside>`, `<footer>`, `<figure>`, `<button>`, `<form>` and `<label>`.

Q2. The box model ●

Every element is a box made of four layers, from inside out:

```

+----- margin -----+   space outside
| +----- border -----+ |
| | +----- padding -----+ | |   space inside
| | |           content           | | |

```

The problem, and the one line fix

By default, `width: 300px` means the **content** is 300px. Add 20px of padding and a 2px border, and the box actually takes 344px on screen. This breaks layouts.

```
* { box-sizing: border-box; }
```

Now `width: 300px` means the whole box is 300px, including padding and border. This is what everyone expects, which is why this line is in almost every CSS reset.

Q3. Flexbox vs Grid — when do you use which? ●

Flexbox works in one direction. A row, or a column.

Grid works in two directions. Rows and columns together.

Use Flexbox for

Navbars, a row of buttons, centring something, spacing items apart, the inside of a card.

```

.navbar {
  display: flex;
  justify-content: space-between; /* spread along the main axis */
  align-items: center;           /* centre on the cross axis */
  gap: 1rem;
}

```

Use Grid for

Page layouts, card galleries, dashboards, anything with real rows and columns.

```

.cards {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(260px, 1fr));
  gap: 1.5rem;
}

```

● That grid line is worth explaining, it sounds senior

Read it as: *"Fit as many columns as you can. Each must be at least 260px wide. Any leftover space is shared equally between them."*

The result is a card grid that goes 4 across on a laptop, 2 on a tablet and 1 on a phone, **with no media queries at all**. Demo that line and explain it.

They work together

A common real layout is Grid for the page, Flexbox inside each card.

Centring, the short version

```
.parent { display: grid; place-items: center; }
```

Q4. Responsive design and mobile first •

What mobile first means

Write the styles for the smallest screen first. Then use `min-width` media queries to **add** complexity for bigger screens.

```
/* base = mobile */
.card { padding: 1rem; }

@media (min-width: 768px) { .card { padding: 2rem; } } /* tablet and up */
@media (min-width: 1024px) { .card { padding: 3rem; } } /* desktop and up */
```

Why not the other way round?

If you write desktop styles first and use `max-width` to shrink them, the phone downloads all the desktop rules and then overrides them. The weakest device does the most work. Mobile first gives the simplest styles to the simplest device.

The essentials

- **The viewport meta tag.** Without this line, nothing responsive works at all:

```
<meta name="viewport" content="width=device-width, initial-scale=1">
```

- **Relative units** instead of fixed pixels: `rem`, `%`, `vw`.
- **Fluid sizing** with `clamp()`:

```
font-size: clamp(1rem, 2.5vw, 2rem); /* min, preferred, max */
```

- **`max-width: 100%` on images** so they never overflow their container.

Q5. Position values

Value	What it does
<code>static</code>	Default. Sits in normal flow.
<code>relative</code>	Moves relative to itself, but keeps its original space.
<code>absolute</code>	Removed from flow. Positioned against the nearest positioned parent.
<code>fixed</code>	Positioned against the viewport. Stays put when scrolling.
<code>sticky</code>	Normal until it reaches a threshold, then behaves like fixed.

`sticky` is what you use for a table header that stays visible while scrolling. It needs a `top` value to work.

Q6. Specificity, in simple terms

When two rules target the same element, the more specific one wins.

```
inline style    = 1000
#id             = 100
.class          = 10
element (div)  = 1
```

`!important` beats everything. Treat it as a warning sign.

"If I need `!important`, it usually means my selectors have got too specific somewhere. I would rather fix the selector than fight it."

Q7. Quick questions

- **`display: none` vs `visibility: hidden` vs `opacity: 0` ?**
Removed from the layout / hidden but still takes up space / fully visible to the layout and **still clickable**.
- **Pseudo class vs pseudo element?** `:hover` is a state. `::before` creates a new sub element.
- **`rem` vs `em` ?** `rem` is relative to the root font size, which is predictable.
`em` is relative to the parent, so it multiplies when nested.
- **CSS variables?** Declare `--brand: #14356b` on `:root`, then use it anywhere as `background: var(--brand)`. They can be changed at runtime with JavaScript, which is how dark mode is usually built.

- **Why is my `z-index` not working?** Two common reasons. It only applies to positioned elements. And a parent with `transform` or `opacity` less than 1 creates a new stacking context that traps its children.
- **What do you think of Tailwind?** Have an opinion but do not be extreme.

"Utility classes keep the styling next to the markup, and unused CSS is removed at build time so the file stays small. The downside is that the JSX gets noisy. I pull repeated patterns into components."

- **CSS Modules?** `styles.card` generates a unique class name, so styles cannot clash between files. Next.js supports them by default.

✓ Check yourself before moving on

1. Flexbox vs Grid, decided in one sentence.
2. Write the `auto-fit` and `minmax` grid line from memory and explain it.
3. Explain mobile first and why `min-width` is better than `max-width`.

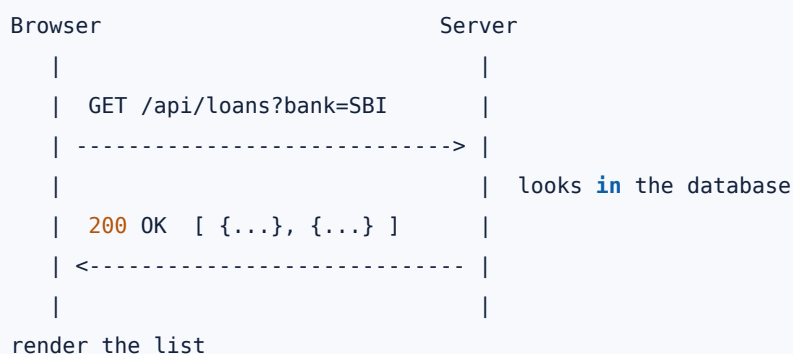
06 · Connecting to REST APIs

Time needed: 30 minutes

The job description says: *"Working closely with our backend team to plug REST APIs and data into the frontend smoothly."* Expect this in the coding round.

First, the context: what an API call actually is

Your React app has no data of its own. The data lives on a server. An API is the agreed way to ask for it.



REST is just a set of conventions for how those requests should look. Use nouns for the address (`/loans`), and a verb to say what you want to do.

Q1. The HTTP basics they may check

Verb	Meaning
GET	Read data. Should never change anything.
POST	Create something new.
PUT	Replace an entire item.
PATCH	Update part of an item.
DELETE	Remove an item.

Status codes

Code	Meaning
200	OK
201	Created

Code	Meaning
400	Bad request. You sent something wrong.
401	Not logged in. "Who are you?"
403	Logged in, but not allowed. "I know who you are, and no."
404	Not found
429	Too many requests, you are rate limited
500	Server error. Not your fault.

- The 401 vs 403 difference is a common quick question. Learn the two sentences.

Q2. `fetch` vs `axios`

	<code>fetch</code>	<code>axios</code>
Setup	Built into the browser	A library, about 13 KB
Error on 404 or 500	No. You must check yourself.	Yes, it throws automatically
JSON	You call <code>res.json()</code>	Automatic both ways
Timeout	Not built in	Built in option
Interceptors	No	Yes

What is an interceptor? A function that runs on every request or every response. It lets you write something once instead of in fifty places.

```
// attach the token to every request, in one place
axios.interceptors.request.use(config => {
  config.headers.Authorization = `Bearer ${getToken()}`;
  return config;
});

// handle every expired session, in one place
axios.interceptors.response.use(
  res => res,
  err => {
    if (err.response?.status === 401) redirectToLogin();
    return Promise.reject(err);
  }
);
```

"For a small project `fetch` is enough. On a bigger codebase I prefer axios for the interceptors, so the auth header and the 401 handling live in one place instead of being repeated in every call."

Q3. The four states ●●

This is the most important idea in this file.

Every screen that loads data has four possible states:

1. **Loading** — the request is in flight. Show a spinner or a skeleton.
2. **Error** — something failed. Show a message and a retry button.
3. **Empty** — it worked, but there is nothing to show. Show "No loans match your filters."
4. **Success** — show the data.

Most candidates only handle loading and success. Saying all four out loud is a genuine hiring signal, because empty and error are what users actually hit.

The full pattern

```
function LoanList() {
  const [loans, setLoans] = useState([]);
  const [loading, setLoading] = useState(true);
  const [error, setError] = useState(null);

  useEffect(() => {
    const controller = new AbortController();

    async function load() {
      try {
        setLoading(true);
        setError(null);

        const res = await fetch('/api/loans', { signal: controller.signal });

        // fetch does NOT throw on 404 or 500, so check it yourself
        if (!res.ok) throw new Error(`Request failed: ${res.status}`);

        setLoans(await res.json());
      } catch (err) {
        // ignore the error we caused ourselves by cancelling
        if (err.name !== 'AbortError') setError(err.message);
      } finally {
        setLoading(false); // always runs, success or failure
      }
    }

    load();

    return () => controller.abort(); // cleanup: cancel if we unmount
  }, []);

  if (loading) return <Spinner />;
  if (error) return <ErrorState message={error} onRetry={load} />;
  if (!loans.length) return <p>No loans match your filters.</p>;

  return <ul>{loans.map(l => <LoanCard key={l.id} loan={l} />)}</ul>;
}
```

Point out these three things without being asked

1. **res.ok** because **fetch** does not throw on a bad status.
2. **AbortController** **in the cleanup** for two reasons, explained next.
3. **finally** so loading always stops, even when the request fails.

Q4. Race conditions, and why `AbortController` matters •

The problem, with a real example

The user types in a search box.

```
t=0ms    types "ax"    → request A starts
t=200ms   types "axis" → request B starts
t=600ms   request B finishes → screen shows results for "axis" <span class="tick">✓</span>
t=900ms   request A finishes → screen shows results for "ax"    ✗ WRONG
```

The older, slower request finished last and overwrote the correct results. The search box now shows results that do not match what is typed.

The fixes

1. **Cancel the old request** with `AbortController`. This is the standard fix.
2. **Debounce** so fewer requests are made in the first place.
3. Use **React Query**, which handles this for you.

`AbortController` in the cleanup solves two problems at once: it prevents this race condition, and it stops React warning that you updated state on a component that no longer exists.

Q5. CORS — you will be asked at least what it is

What it is

CORS is a **browser** security rule. A page loaded from `site-a.com` is not allowed to read a response from `site-b.com`, unless `site-b.com` explicitly permits it with a response header:

```
Access-Control-Allow-Origin: https://site-a.com
```

The two things to say

1. **It is enforced by the browser, not the server.** This is why the exact same request works fine in Postman but fails in the browser.
2. **It is fixed on the backend, not the frontend.** The server must send the header. A dev proxy is a local workaround, not a real fix.

For requests that are not simple, the browser first sends an `OPTIONS` request to ask permission. That is called a **preflight**.

Q6. React Query and SWR — worth mentioning

"A lot of what people call global state is really just a cached copy of server data. React Query handles that properly: caching, background refetching, deduplicating identical requests, retries, and loading and error flags. It replaces most of the `useEffect` plus `useState` code above, including the race condition handling."

```
const { data, isLoading, error } = useQuery({
  queryKey: ['loans'],
  queryFn: getLoans,
});
```

Q7. Authentication on the frontend

- Store the token in an **httpOnly cookie** rather than `localStorage`. `httpOnly` means JavaScript cannot read it, so an injected script cannot steal it.
- Attach it with an interceptor so it is written once.
- Handle **401** in one place: clear the session and send the user to login.
- **Never put secrets in frontend code.** In Next.js, anything named `NEXT_PUBLIC_*` is visible in the browser. If you need to use a secret key, call it from an API route on the server instead.

Q8. "How do you work with the backend team?" — the job description's real question

Answer as a teammate, not just a coder:

"I ask for the contract early: the endpoint, the request shape, the response shape and the error shape. Agreeing on that before either side builds saves a lot of rework.

If the API is not ready, I mock the response so I am not blocked, and swap the URL in later.

I check pagination and the error format up front, because those are usually what break during integration.

And if a response shape does not fit the UI well, I would rather raise it than quietly write conversion code in five different components."

✓ Check yourself before moving on

1. Name the four states, in order.
2. Explain the race condition example, and how `AbortController` fixes it.
3. Explain CORS in two sentences, including whose job it is to fix.
4. What does `fetch` do on a 404?

07 · Machine Coding Round

Time needed: 60 minutes. And you must TYPE these, not read them.

Reading code makes you feel prepared. It does not make you prepared. Open CodeSandbox or StackBlitz and build problems 1 and 2 from an empty file. That is how you should spend this hour.

| First, the context: what they are actually grading

They are not checking whether your code is beautiful. In 45 minutes nobody writes beautiful code. They are checking six things:

1. Can you break the problem into state and components?
2. Do you know React well enough to not fight it?
3. Do you handle the cases that are not the happy path?
4. Do you talk while you work?
5. Do you notice your own mistakes?
6. Would it be pleasant to sit next to you for a day?

| How to run the round

1. Clarify for 60 seconds. Do not start typing immediately.

"Should the filtering happen on the client, or should I call the API each time?"
"Roughly how many items are we dealing with?"
"Should the state survive a refresh?"

Candidates who ask questions score higher than candidates who start typing. Every time.

2. Say your plan out loud before you write it.

"I will keep the typed text in state, debounce it, then filter the list while rendering. Then I will handle the empty case."

3. Build the ugly working version first. Get it working, then improve it. Do not style anything until the logic works.

4. Keep talking while you type. Silence makes the interviewer think you are stuck. Narrate what you are doing, even if it feels awkward.

5. Say the edge cases out loud, even the ones you do not have time to build. *"I would add an error state here if I had more time"* still earns you the point.

6. Refactor at the end if there is time. Pull something into a custom hook, or split a component. It shows you know what good code looks like.

Common problems at this level: search and filter a list, todo CRUD, a counter, an accordion or tabs, a star rating, form validation, fetching and displaying users, pagination, a modal, a countdown timer.

| Problem 1 ● — Search and filter a list, with debounce

The most commonly asked machine coding question in Indian frontend interviews.

What they are testing: derived state, debouncing, keys, and whether you remember the empty state.

```

import { useState, useEffect, useMemo } from 'react';

function useDebounce(value, delay = 400) {
  const [debounced, setDebounced] = useState(value);
  useEffect(() => {
    const t = setTimeout(() => setDebounced(value), delay);
    return () => clearTimeout(t);
  }, [value, delay]);
  return debounced;
}

export default function BankSearch({ banks }) {
  const [query, setQuery] = useState('');
  const [type, setType] = useState('all');
  const debouncedQuery = useDebounce(query);

  // derived state - NOT useState + useEffect
  const filtered = useMemo(() => {
    return banks.filter(b => {
      const matchesQuery = b.name.toLowerCase().includes(debouncedQuery.toLowerCase().trim());
      const matchesType = type === 'all' || b.type === type;
      return matchesQuery && matchesType;
    });
  }, [banks, debouncedQuery, type]);

  return (
    <div>
      <input
        value={query}
        onChange={e => setQuery(e.target.value)}
        placeholder="Search banks..."
        aria-label="Search banks"
      />
      <select value={type} onChange={e => setType(e.target.value)}>
        <option value="all">All</option>
        <option value="public">Public</option>
        <option value="private">Private</option>
        <option value="nbfc">NBFC</option>
      </select>

      {filtered.length === 0
        ? <p>No banks match "{debouncedQuery}"</p>
        : <ul>{filtered.map(b => <li key={b.id}>{b.name} - {b.rate}%</li>)}</ul>}
    </div>
  );
}

```

Explain these four things while you type

1. The filtered list is calculated during render, not stored in state.

"I could have put `filtered` in state and updated it in a `useEffect`, but then I have two sources of truth that can go out of sync, plus an extra render. It is derived data, so I calculate it."

2. The debounce reduces work.

"Without this, typing nine characters runs the filter nine times. If it were an API call, that would be nine requests instead of one."

3. The key is a stable id, never the array index.

4. The edge cases are handled. `.trim()` removes accidental spaces, and `.toLowerCase()` on both sides makes the search case insensitive. And when nothing matches, the user sees a message instead of a blank area.

Problem 2 ● — Fetch + loading / error / empty / retry

```
import { useState, useEffect, useCallback } from 'react';

export default function Users() {
  const [users, setUsers] = useState([]);
  const [loading, setLoading] = useState(true);
  const [error, setError] = useState(null);

  const load = useCallback(async (signal) => {
    try {
      setLoading(true);
      setError(null);
      const res = await fetch('https://jsonplaceholder.typicode.com/users', { signal });
      if (!res.ok) throw new Error(`HTTP ${res.status}`);
      setUsers(await res.json());
    } catch (e) {
      if (e.name !== 'AbortError') setError(e.message);
    } finally {
      setLoading(false);
    }
  }, []);

  useEffect(() => {
    const c = new AbortController();
    load(c.signal);
    return () => c.abort();
  }, [load]);

  if (loading) return <p>Loading...</p>;
  if (error) return (
    <div role="alert">
      <p>Couldn't load users: {error}</p>
      <button onClick={() => load()}>Retry</button>
    </div>
  );
  if (!users.length) return <p>No users found.</p>;

  return <ul>{users.map(u => <li key={u.id}>{u.name}</li>)}</ul>;
}
```

Why this scores well

- All four states are handled: loading, error, empty, and success.
- **The retry button.** Almost nobody adds this. An error state the user cannot escape from is a dead end.
- `res.ok` is checked, because `fetch` does not throw on a 404 or 500.

- The request is cancelled on unmount with `AbortController`.
- `finally` means loading always stops, even when the request fails.

Problem 3 — Todo list (CRUD + no mutation)

```
export default function Todos() {
  const [todos, setTodos] = useState([]);
  const [text, setText] = useState('');

  const add = (e) => {
    e.preventDefault();
    const value = text.trim();
    if (!value) return; // edge case: empty input
    setTodos(t => [...t, { id: crypto.randomUUID(), text: value, done: false }]);
    setText('');
  };

  const toggle = (id) =>
    setTodos(t => t.map(todo => todo.id === id ? { ...todo, done: !todo.done } : todo));

  const remove = (id) => setTodos(t => t.filter(todo => todo.id !== id));

  return (
    <>
      <form onSubmit={add}>
        <input value={text} onChange={e => setText(e.target.value)} />
        <button type="submit">Add</button>
      </form>
      <ul>
        {todos.map(todo => (
          <li key={todo.id}>
            <input type="checkbox" checked={todo.done} onChange={() => toggle(todo.id)} />
            <span style={{ textDecoration: todo.done ? 'line-through' : 'none' }}>{todo.text}</span>
            <button onClick={() => remove(todo.id)}>x</button>
          </li>
        ))}
      </ul>
      <p>{todos.filter(t => !t.done).length} remaining</p>
    </>
  );
}
```

Details worth pointing out

- `<form onSubmit>` instead of a button `onClick`. This means the Enter key works, which is what users expect. Interviewers notice this.

- **Functional updaters** (`setTodos(t => ...)`) so you always work from the latest state.
 - `map` and `filter` **never change the original array**. They return a new one, which is exactly what React needs to detect a change.
 - `crypto.randomUUID()` **for the id**, not the array index.
 - **The remaining count is calculated, not stored** in another piece of state.
-

Problem 4 — Form with validation

```
export default function LeadForm() {
  const [values, setValues] = useState({ name: '', email: '', phone: '' });
  const [errors, setErrors] = useState({});
  const [touched, setTouched] = useState({});
  const [status, setStatus] = useState('idle'); // idle | submitting | success | error

  const validate = (v) => {
    const e = {};
    if (!v.name.trim()) e.name = 'Name is required';
    if (!/^[^\s@]+@[^\s@]+\.[^\s@]+$/.test(v.email)) e.email = 'Enter a valid email';
    if (!/^[6-9]\d{9}$/.test(v.phone)) e.phone = 'Enter a valid 10-digit mobile number';
    return e;
  };

  const handleChange = (e) => {
    const next = { ...values, [e.target.name]: e.target.value };
    setValues(next);
    if (touched[e.target.name]) setErrors(validate(next)); // re-validate only after blur
  };

  const handleBlur = (e) => {
    setTouched(t => ({ ...t, [e.target.name]: true }));
    setErrors(validate(values));
  };

  const handleSubmit = async (e) => {
    e.preventDefault();
    const found = validate(values);
    setErrors(found);
    setTouched({ name: true, email: true, phone: true });
    if (Object.keys(found).length) return;
    setStatus('submitting');
    try {
      await fetch('/api/leads', { method: 'POST', body: JSON.stringify(values) });
      setStatus('success');
    } catch { setStatus('error'); }
  };

  return (
    <form onSubmit={handleSubmit} noValidate>
      {[ 'name', 'email', 'phone' ].map(field => (
        <div key={field}>
          <label htmlFor={field}>{field}</label>
          <input id={field} name={field} value={values[field]}
            onChange={handleChange} onBlur={handleBlur}
            aria-invalid={!!errors[field]} />
          {touched[field] && errors[field] && <small role="alert">{errors[field]}</small>}
        </div>
      )]}
    </form>
  );
}
```

```

    </div>
  )}
  <button disabled={status === 'submitting'}>
    {status === 'submitting' ? 'Submitting...' : 'Submit'}
  </button>
  {status === 'success' && <p>Thanks – we'll call you shortly.</p>}
</form>
);
}

```

The three details that make this look experienced

1. Errors only appear after the user leaves a field (`onBlur`).

Showing "Enter a valid email" while someone is still typing the first letter of their email is bad UX. That is what the `touched` state is for.

2. The submit button is disabled while the request is in flight.

Otherwise an impatient user clicks three times and creates three applications.

3. One `handleChange` for all fields, using `e.target.name`. Writing a separate handler for every field does not scale.

Problem 5 — Accordion (controlled, one open at a time)

```

function Accordion({ items }) {
  const [openId, setOpenId] = useState(null);
  return (
    <div>
      {items.map(item => {
        const isOpen = openId === item.id;
        return (
          <div key={item.id}>
            <button aria-expanded={isOpen}
              onClick={() => setOpenId(isOpen ? null : item.id)}>
              {item.title} {isOpen ? '-' : '+'}
            </button>
            {isOpen && <p>{item.body}</p>}
          </div>
        );
      })}
    </div>
  );
}

```

Why one `openId` instead of a boolean per item

If each item had its own `isOpen` boolean, you would have to remember to close all the others every time one opens. It is easy to get wrong. With a single `openId`,

"only one open at a time" is guaranteed by the shape of the state itself.

If they ask for multiple open at once, hold a `Set` of ids instead.

`aria-expanded` tells a screen reader whether the section is open. And if this is an FAQ, you can mention that adding FAQ structured data would make it eligible for a rich result in Google.

Problem 6 — Star rating

```
function Rating({ value = 0, onChange, max = 5 }) {
  const [hover, setHover] = useState(0);
  return (
    <div onMouseLeave={() => setHover(0)}>
      {Array.from({ length: max }, (_, i) => i + 1).map(star => (
        <button key={star} type="button"
          aria-label={`Rate ${star} of ${max}`}
          onMouseEnter={() => setHover(star)}
          onClick={() => onChange(star)}>
          {star <= (hover || value) ? '★' : '☆'}
        </button>
      ))}
    </div>
  );
}
```

The trick is `hover || value`

While the mouse is over star 4, `hover` is 4, so stars 1 to 4 are filled as a preview. When the mouse leaves, `hover` becomes 0, which is falsy, so it falls back to the actual saved `value`. One line handles both the preview and the real state.

Problem 7 — Pagination (client-side)

```
function Paginated({ items, perPage = 10 }) {
  const [page, setPage] = useState(1);
  const totalPages = Math.ceil(items.length / perPage);
  const slice = items.slice((page - 1) * perPage, page * perPage);

  useEffect(() => { setPage(1); }, [items]); // reset when the data set changes

  return (
    <>
      <ul>{slice.map(i => <li key={i.id}>{i.name}</li>)}</ul>
      <button disabled={page === 1} onClick={() => setPage(p => p - 1)}>Prev</button>
      <span>Page {page} of {totalPages} | 1</span>
      <button disabled={page >= totalPages} onClick={() => setPage(p => p + 1)}>Next</button>
    </>
  );
}
```

Edge cases to name out loud

- Disable Previous on page 1 and Next on the last page.
- An empty list should not show "Page 1 of 0".
- **Reset to page 1 when the data changes.** If the user is on page 5 and applies a filter that leaves 3 results, they would see an empty page. This is a very common bug.

(You have built `xpagination` before. Mention it.)

Problem 8 — Counter / timer (the `useRef` + cleanup check)

```
function Timer() {
  const [seconds, setSeconds] = useState(0);
  const [running, setRunning] = useState(false);
  const idRef = useRef(null);

  useEffect(() => {
    if (!running) return;
    idRef.current = setInterval(() => setSeconds(s => s + 1), 1000); // functional updater!
    return () => clearInterval(idRef.current); // cleanup!
  }, [running]);

  return (
    <>
      <h1>{seconds}</h1>
      <button onClick={() => setRunning(r => !r)}>{running ? 'Pause' : 'Start'}</button>
      <button onClick={() => { setRunning(false); setSeconds(0); }}>Reset</button>
    </>
  );
}
```

They are checking exactly two things here

1. `setSeconds(s => s + 1)` and not `setSeconds(seconds + 1)` .

The interval callback was created once. It captured `seconds` as 0 in a closure and will never see a newer value. So `seconds + 1` would be `0 + 1` forever. The function form always receives the latest value.

2. The `clearInterval` cleanup.

Without it, the timer keeps running after the component is gone and tries to update state that no longer exists.

Point both out while you type. They are the entire reason this problem is asked.

If they ask for plain JS/DOM instead

```
document.querySelector('#list').addEventListener('click', (e) => { // delegation
  const btn = e.target.closest('button[data-id]');
  if (!btn) return;
  removeItem(btn.dataset.id);
});
```

✓ Tonight's task

Build **Problem 1** and **Problem 2** from an empty file, with no copy and paste.

If you can build those two while explaining what you are doing, you can handle anything they are likely to ask at this level.

08 · Frontend System Design

Time needed: 45 minutes

This section was missing from the first version of the guide. It matters, because at this level the system design round is where they check if you can **plan** before you code, not just type React.

| First, the context: what "frontend system design" actually means

When a backend engineer hears "system design", they think about databases, servers and load balancing. **You will not be asked that.**

A frontend system design question sounds like this:

- "How would you build the loan search page?"
- "Design an autocomplete search box."
- "How would you structure a multi step application form?"
- "We have a page with 5,000 rows. How do you make it fast?"

They are checking six things:

1. Do you **ask questions** before you start?
2. Can you break a screen into **components**?
3. Do you know **where the state should live**?
4. Do you think about the **API**, not just the UI?
5. Do you handle **loading, error and empty** cases?
6. Do you think about **performance** without being asked?

You do not need a perfect answer. You need a clear, ordered way of thinking.

| The framework: answer every design question in this order

Memorise these six steps. Use them for any question they ask.

- | | |
|----------------|---|
| 1. CLARIFY | Ask 2-3 questions. Never start immediately. |
| 2. COMPONENTS | Break the screen into a component tree. |
| 3. STATE | Decide what state exists and who owns it. |
| 4. DATA | What API calls, when, and in what shape. |
| 5. EDGE CASES | Loading, error, empty, slow network, long text. |
| 6. PERFORMANCE | One or two improvements, with a reason. |

Step 1 is the one candidates skip, and it is the one that scores most.

Starting to code before asking anything is the most common failure in this round.

Useful clarifying questions that work for almost any problem:

- "Is the filtering done on the client, or does the API do it?"
- "Roughly how many items are we showing? Ten, or ten thousand?"
- "Does this need to work for logged out users, and does it need to rank on Google?"
- "Is the data the same for everyone, or is it per user?"

That last question is powerful in a Next.js interview, because the answer decides SSG, ISR or SSR.

Design 1 • — An autocomplete search box

The most commonly asked frontend design question in India.

Step 1: Clarify

- "Does the API return results, or do I filter a list I already have?"
- "Should recent searches be remembered?"
- "Do we need keyboard navigation with arrow keys?"

Step 2: Components

```
SearchBox
├─ <input>           the text field
├─ SuggestionList    shown only when there are results
│   └─ SuggestionItem
└─ StatusArea        spinner / "no results" / error
```

Step 3: State

State	Where it lives	Why
query	SearchBox	what the user typed
debouncedQuery	SearchBox (custom hook)	what we actually search with
results	SearchBox	what came back
loading , error	SearchBox	to show the right status
activeIndex	SearchBox	which suggestion is highlighted

Step 4: Data flow

```
user types → query updates on every keystroke
           → debounce waits 400ms of silence
           → ONE API call fires
           → results render
```

Step 5: Edge cases — say all of these out loud

- **Empty query.** Do not call the API at all. Close the dropdown.
- **No results.** Show "No banks found for axis". Do not show an empty box.
- **Error.** Show a short message and let them try again.
- **Race condition.** The user types "ax", then "axis". Two requests are running. If "ax" answers last, the screen shows the wrong list. Fix it by cancelling the old request with `AbortController`.
- **Clicking outside** should close the dropdown.
- **Keyboard.** Arrow keys move, Enter selects, Escape closes.

Step 6: Performance

- **Debounce** turns 9 keystrokes into 1 request. That is a 90% reduction in load on the backend.
- **Cache** results per query in a simple object, so pressing backspace does not refetch something you already have.
- If the list is very long, **virtualise** it so only visible rows are in the DOM.

The 60 second version to say out loud

"I would hold the typed text in state, and pass it through a `useDebounce` hook so the API is only called after the user pauses for around 400 milliseconds. The request result goes into `results`, with separate `loading` and `error` state.

I would show four states: loading, error, empty and results, because an empty dropdown with no message looks broken.

The main bug to guard against is a race condition. If two requests are in flight, the slower older one can overwrite the newer results. I cancel the previous request with an `AbortController` in the effect cleanup. And I would cache results per query so backspacing does not refetch."

Design 2 ● — A loan listing page with filters and pagination

This is essentially their real product, so it is very likely to come up.

Step 1: Clarify

- "How many loans are there in total? Hundreds or thousands?"
- "Should filters appear in the URL, so a student can share the link?"
- "Does this page need to rank on Google?"

That last question changes the whole design, so ask it.

Step 2: Components

```
LoansPage
├─ FilterPanel
│   ├─ BankTypeFilter (public / private / NBFC)
│   ├─ AmountRangeFilter
│   └─ ClearAllButton
├─ ResultsHeader    "24 loans found" + sort dropdown
├─ LoanList
│   └─ LoanCard × N
└─ Pagination
```

Step 3: State, and the important decision

```
filters = { type: 'public', maxRate: 10, page: 2 }
loans   = []
loading / error
```

The key insight to mention: keep the filters in the URL.

```
/loans?type=public&maxRate=10&page=2
```

Why this matters:

- The student can **share or bookmark** their filtered view.
- The **back button works** correctly.
- Refreshing the page keeps the filters.
- And for Next.js, the server can read those filters and render the page on the server, so it is crawlable.

If you keep filters only in `useState`, all four of those break. Saying this shows product thinking, not just React knowledge.

Step 4: Data — client side or server side filtering?

	Filter on client	Filter on server
Good when	Under ~500 items	Thousands of items

	Filter on client	Filter on server
Speed	Instant, no network	One request per change
Data sent	All items at once	Only the current page

"With a few hundred loans I would fetch once and filter in memory, which feels instant. With thousands, I would send the filters to the API and paginate on the server, because downloading everything would be slow and wasteful on mobile data."

Step 5: Edge cases

- No results after filtering → "No loans match your filters" **plus a Clear filters button**. Do not leave a dead end.
- Changing a filter must **reset the page back to 1**. A very common bug.
- Disable Previous on page 1 and Next on the last page.
- Long bank names must not break the card layout.

Step 6: Performance and rendering strategy

"The loan list is the same for every visitor, so I would use **ISR** with a revalidate of about an hour. That gives static file speed from the CDN, and the rates are never more than an hour stale. Images go through `next/image`. If the list ever grows into thousands of rows on one page, I would virtualise it."

Design 3 ● — A multi step application form

This is your home ground. You built a 14 screen onboarding flow (Business Card) and a 7 screen recall workflow with navigation guards. If this question comes up, answer it from real experience, not theory.

Step 1: Clarify

- "How many steps, and can the user go back?"
- "If they close the browser, should the progress be saved?"
- "Is validation per step, or all at the end?"

Step 2: Components

```
FormWizard      ← owns ALL the form data
├─ ProgressBar  step 2 of 4
├─ StepPersonal
├─ StepEducation
├─ StepLoanAmount
├─ StepReview
└─ NavButtons    Back / Next / Submit
```

Step 3: State — the important decision

All form data lives in the parent `FormWizard`, not in each step.

```
const [formData, setFormData] = useState({ name: '', email: '', course: '' });
const [step, setStep] = useState(1);
const [errors, setErrors] = useState({});
```

Why? If each step kept its own state, going back to step 1 would unmount step 2 and destroy everything typed there. Keeping the data in the parent means the steps become simple display components, and nothing is lost when moving between them.

(This is exactly the bug you fixed on the Business Card application at Standard Bank. Say so. A design answer backed by "I hit this in production and here is what went wrong" is far stronger than a theoretical one.)

Step 4 and 5: Validation and edge cases

- Validate the current step before allowing Next.
- Show errors only **after** the user leaves a field, not while they are typing the first letter.
- Disable Submit while the request is in flight, so it cannot be double submitted.
- Save progress to `localStorage` so a refresh does not lose everything.
- On submit failure, keep all the data and show a retry. Never clear the form.

Step 6: Performance

"The steps that are not visible do not need to be rendered at all. If a step imported something heavy, like a document upload widget, I would lazy load it with `next/dynamic` so it is not in the initial bundle."

Design 4 — "How would you make a reusable component?"

They may hand you a button or a card and ask how you would design its API.

The principles, in simple words

1. It should not know where it is used.

A `LoanCard` should not contain `if (page === 'home')`. Pass differences in as props.

2. Props describe *what*, not *how*.

```
<Button variant="primary" size="sm" loading={isSaving}>Apply</Button>
```

Not `<Button backgroundColor="#0055ff" paddingLeft="12px">`. If you pass raw styles, every screen looks slightly different and you have no design system.

3. Use `children` for the content.

```
<Card>
  <h3>SBI</h3>
  <p>9.15%</p>
</Card>
```

This is far more flexible than `<Card title="SBI" subtitle="9.15%" />`, because the next screen will always need something you did not predict.

4. Pass the rest of the props through.

```
function Button({ variant, children, ...rest }) {
  return <button className={styles[variant]} {...rest}>{children}</button>;
}
```

Now `onClick`, `disabled`, `type` and `aria-label` all work without you adding them one by one.

5. Handle its own states. A button should support normal, hover, disabled and loading. A card should survive a very long title.

The warning sign to mention

"If a component has grown to fifteen boolean props like `isHomePage`, `isCompact`, `hideFooter`, that usually means it is really two different components. I would split it rather than keep adding flags."

Design 5 — How would you structure a large Next.js project?

```
app/                                routes only
  layout.js
  page.js
  loans/
    page.js
    [id]/page.js
  api/
components/
  ui/                               generic, reusable: Button, Card, Input, Modal
  loans/                           feature specific: LoanCard, FilterPanel
  layout/                          Header, Footer, Sidebar
hooks/                             useDebounce, useFetch, useLocalStorage
lib/                               api client, formatters, constants
styles/
public/                           images, fonts, icons
```

The one rule worth stating:

"I separate **generic UI components** from **feature components**. Anything in `components/ui` knows nothing about loans and could be copied into another project. Anything in `components/loans` is allowed to know about our data. That boundary stops the shared components from slowly filling up with business logic."

Design 6 — "This page is slow. How do you fix it?"

Never jump to a solution. Show a process. (Full detail is in file 04.)

1. MEASURE Lighthouse + DevTools Network and Performance tabs.
Throttle to Slow 4G. "I do not optimise what I have not measured."
2. CATEGORISE Which **is** it?
 - Too many / too large downloads → images, bundle
 - Slow server response (TTFB) → caching, rendering strategy
 - Heavy JavaScript blocking → long tasks, re-renders
3. FIX THE BIGGEST ONE FIRST Usually images, then bundle size.
4. MEASURE AGAIN Prove the change worked.

Then name concrete fixes: `next/image` for format and size, code splitting per route, removing heavy libraries, virtualising long lists, debouncing inputs, and moving pages to SSG or ISR so the HTML is ready before JavaScript loads.

Design 7 — How do you handle authentication on the frontend?

The simple flow

1. User submits email + password
2. Server checks them, **returns** a **token**
3. **Token** is stored ← the important decision
4. Every later request sends the **token**
5. Server **returns** **401** when it expires → send user back **to** login

The decision to explain

"I would keep the token in an **httpOnly cookie** rather than `localStorage`. `httpOnly` means JavaScript cannot read it, so if someone manages to inject a script into the page, they still cannot steal the session. Anything in `localStorage` is readable by any script running on the page. For a product handling student financial data, that difference matters."

The frontend pieces

- An **axios interceptor** to attach the token to every request, in one place.
- A second interceptor to catch **401** responses and redirect to login, in one place.
- **Protected routes.** In Next.js, middleware can redirect before the page even renders.
- An `AuthContext` so any component can read the current user.

Design 8 — A toast / notification system

A nice small design question, and it shows you understand context.

The problem

Any component anywhere might need to say "Saved successfully". You cannot pass a callback down to all of them.

The design

```
ToastProvider (context, sits near the root)
├─ holds: toasts = [{ id, message, type }]
├─ gives: showToast(message, type)
└─ renders: <ToastContainer> fixed in a corner
```

```
// any component, at any depth
const { showToast } = useToast();
showToast('Application submitted', 'success');
```

Edge cases to name: auto dismiss after a few seconds, allow manual close, stack multiple toasts instead of overwriting, and clear the timer on unmount so it does not fire after the component is gone.

| The three sentences that work in any design answer

Keep these ready. They fit almost every question:

1. **"Before I start, can I ask two things about the requirements?"**
 2. **"Every data driven screen has four states: loading, error, empty and success. Empty is the one people forget."**
 3. **"Since this is a search driven business, I would check whether this page needs to be crawlable, because that decides whether it is client rendered or server rendered."**
-

| ✓ Check yourself before moving on

1. Say the six step framework from memory: clarify, components, state, data, edge cases, performance.
2. Give the autocomplete answer out loud in about 60 seconds.
3. Explain why filters should live in the URL, giving three reasons.
4. Explain client side vs server side filtering, and when you would use each.

09 · HR, Behavioural & the Offer Conversation

Time needed: 45 minutes. Say every answer OUT LOUD.

Reading these silently is worth about a fifth of saying them. Your mouth needs the practice, not your eyes.

1. • "Tell me about yourself" and the four hard questions

Your full script for this is in file 10, section "Tell me about yourself".

It is built from your real Standard Bank experience, so use that version.

File 10 also has prepared answers for the four questions your resume creates, and you must have all four ready:

1. **"Your last role ended in March. What have you been doing since?"**
2. **"Why did you leave Zensar?"**
3. **"You have Java and Spring Boot. Why a pure frontend role?"**
4. **"How much Next.js have you actually used?"**

Do not walk into the room without solid answers to those. They are the most predictable questions you will face tomorrow.

Rules for the opening answer: under two minutes, no life story starting from school, and finish on *why this company*. Say it twice tonight and twice in the morning.

2. • "Why WeMakeScholars?" — do the homework, it's 3 sentences

Use their own facts back at them:

"Three reasons. It's a 10-year-old, 250-person organisation, so it's stable but still building — I'd rather own real features than maintain someone else's legacy code. Second, the product is free for students and funded under Digital India, so the frontend genuinely decides whether someone finds and completes a loan application — the work has a visible outcome. Third, it's a search-driven product, which means Next.js, SSR and Core Web Vitals aren't optional here. That's the skill set I want to grow in."

Never say: "I need a job", "I want to learn", "the package is good".

Do mention you know they were inaugurated by Dr. Shashi Tharoor / recognised by Forbes / won the ET award — once, lightly. It shows you read.

3. • The filter questions — answer instantly and without hedging

These decide whether you move forward. Hesitation here reads as "will leave in six months", which is precisely what the 18-month clause exists to prevent.

"We need an 18-month commitment. Are you comfortable?"

"Yes. I'm looking for a place to build depth, not to switch in a year — 18 months is about the minimum time to actually own something end to end. I'd just like to understand how it's documented in the offer."

(Asking that follow-up is not a red flag — it's professional. Do read the clause before you sign: ask whether it's a written commitment or a bond with a financial penalty, and whether any amount is recoverable if you leave early. Ask it calmly at offer stage, not in round one.)

"What's your notice period?" → This is an advantage for you. Use it.

Your Zensar role ended in March, so you are an immediate joiner. Most candidates they interview will be serving 60 or 90 days.

"I'm available immediately. I can join as soon as you need me to."

Say it plainly and with a slight lift. For a hiring manager who needs someone productive this quarter, "immediate" is worth real money and sometimes wins the offer on its own. *(The 2 month notice in the JD is what you would serve at WeMakeScholars once employed. It is not a question about you now.)*

"Are you okay with work from office in Hyderabad?" → "Yes." Full stop. Don't open a negotiation you weren't asked to have.

"1st and 3rd Saturdays are working." → "That's fine — I saw it in the JD, and it makes sense given you work with banks."

"Timings are 10-11 AM login, 8 hours." → "Works for me."

⚠ If any of these are genuinely a problem for you, say so **now**, politely. Accepting and renegotiating later burns the relationship and your reference.

4. • Salary — the band is 4 to 6 LPA, and you should be aiming at the top of it

Your position, honestly

You have about 1 year 4 months of production experience on a real banking platform, plus an internship, plus full stack ability. **That profile sits at the top of a 4 to 6 LPA band, not the middle.** Candidates with only college projects are the ones who get placed at 4.

So your target is **6**, and your job in the technical rounds is to earn it before the number is ever discussed.

The answer to give

"I saw the range in the job description and I'm comfortable working within it. Given that I've been shipping production React and TypeScript on a banking platform for the last year and a half, including the performance work I mentioned, I'd be looking at the upper end of that band. But I'm more interested in the role than in a specific number, and I'm sure we can agree something if the fit is right."

If they ask for your previous CTC

Answer honestly with the real figure. Payslip verification is routine, and a lie found later costs you the offer after you have already resigned from somewhere else. If your previous CTC was low, do not let it anchor you:

"That was my package as a fresher joining a service company. I'd expect this role to reflect what I can do now rather than what I started on."

One more thing worth knowing

A 4 to 6 LPA band is normally written for someone with little or no production experience. You have a year and a half of it, across React, TypeScript, Redux, Spring Boot and PostgreSQL, on a banking platform. It is reasonable to check whether the band was written with your profile in mind.

You can ask this politely, once, without sounding difficult:

"Can I check whether that range was set for a fresher profile? I'm coming in with about a year and a half of production experience across the full stack, so I wanted to understand how you'd place someone at my level."

That is a fair question and a reasonable employer will answer it. If the answer is that the band is fixed, you still have a decision to make, and it is yours: this is a product company, it is in your city, and it is the frontend specialisation you want. Those can be worth accepting a number for. Just make that choice knowingly rather than by accident.

Your availability is worth something too. Most candidates are serving 60 or 90 days of notice. You can join immediately, which saves them months of waiting. Say so clearly, because hiring managers value it and it is fair to count it in your favour:

"I'm available to join immediately, so there's no notice period to wait out."

Tactics

- **Never name a number below 4.** That is their published floor. Going under it means bidding against yourself.
- **Let them go first if you can.** "What range have you budgeted for this role?" is a completely normal question.
- If you want the top of the band, **earn it in the technical round first**, then cite it: "Based on the Next.js and performance work we discussed, I'd be looking at the upper end."
- Ask **CTC vs in-hand** breakdown at offer stage — how much is fixed, variable, and what's deducted. A 6 LPA CTC and a 6 LPA fixed are different jobs.
- Never lie about your current CTC. Payslip verification is routine.

5. Behavioural questions — answer with STAR

Situation → Task → Action → Result. Keep it under 90 seconds and always land on a result.

"Tell me about a challenging technical problem."

Use the dashboard performance story. It is your strongest.

(Situation) We consolidated four separate modules into one requests page and dashboard for corporate users. **(Task)** After the merge the page felt slow, and Lighthouse was at 62. **(Action)** I measured first with Lighthouse and the DevTools Performance tab rather than guessing. The two real causes were the bundle size, because we had merged four modules into one, and unnecessary re-renders, because our Redux selectors were creating new arrays on every store update. I added code splitting with `React.lazy` and memoised the selectors with `createSelector`. **(Result)** Lighthouse went from 62 to 88, about 35% faster on initial load. What I took from it is that measuring first saves time, because the obvious suspect is often not the real one."

"Tell me about a challenging bug."

(S/T) On the Business Card application, which is fourteen screens, users were losing what they had typed when they navigated back a step. **(A)** The cause was that each step was holding its own local state, so going back unmounted the component and destroyed it. I lifted all the form data into the parent wizard and made each step a controlled presentational component. **(R)** The bug went away and the code got simpler. Adding a new screen became a small change instead of new state logic. It taught me to ask *who should own this state* before writing any of it."

"A time you disagreed with someone."

"We needed an audit trail of who downloaded or emailed each report. The initial suggestion was to add logging code inside each of those methods. I was concerned about touching a lot of working business logic for something that was really a cross-cutting concern. Rather than just objecting, I put together an AOP approach using `@Aspect` pointcuts and showed that it captured the same information without changing a single line of the existing logic. We went with that. I have learned that showing an alternative works much better than arguing about one."

"Tight deadline?" → Talk about scoping: shipping the core flow properly and flagging what you cut, rather than shipping everything half-done. Managers hire people who communicate trade-offs early.

"Your weakness?" → Real, but bounded, plus the fix:

"I used to over-polish UI details before the feature was fully working. I've started building the working version first and reviewing the polish at the end — it keeps me from spending an hour on a hover state that gets redesigned anyway." *(Never "I'm a perfectionist" or "I work too hard". Both read as rehearsed.)*

"Where do you see yourself in 3 years?" → "Owning a significant part of a frontend codebase — the person who's the reference point for performance and component architecture on the team." *(Aligns with 18 months. Don't say "starting my own company" or "moving abroad".)*

"How do you work with designers?" → "I go through the design early and ask about the states the mock doesn't show — loading, error, empty, long text, small screens. Catching those in a 10-minute conversation is far cheaper than in QA."

"How do you keep code maintainable?" → "Small components with one job, clear naming, no magic numbers, shared logic in custom hooks, and a folder structure someone new can navigate. My test is whether a teammate can change it without asking me how it works."

6. • Questions YOU ask them — *never say "no, I'm good"*

Have 4 ready; ask 2–3. Ending with no questions reads as no interest.

Technical (ask these in the tech round):

1. "Is the frontend on the App Router or Pages Router, and are you migrating?"
2. "How do you currently handle rendering strategy for the public pages — mostly SSG/ISR, or SSR?"
3. "Do you track Core Web Vitals in production, and is there a performance budget?"
4. "What does the frontend-to-backend workflow look like — do you get API contracts up front?"
5. "Is there a shared component library, or does each page build its own?"

Role / team (ask in the manager or HR round):

6. "What would a successful first three months in this role look like?"

7. "How big is the frontend team, and would I own features end to end?"
8. "Is there code review, and who reviews frontend work?"

The one that makes you memorable — ask it in the manager round:

"What's the biggest frontend problem you're hoping this hire helps solve?"

Then **listen**, and connect your answer to it. That's how you close.

7. Logistics for tomorrow

- Laptop charged + charger · backup mobile hotspot · water · pen and notebook.
 - 2 printed copies of your resume, plus a copy open on your laptop.
 - Your GitHub open in a tab — and **your best 2 projects running**, so "can you show me?" gets a yes in five seconds, not a five-minute build.
 - Reach 15 minutes early. If it's online, join 5 minutes early and test camera/mic.
 - Dress: smart casual — collared shirt. Slightly overdressed beats underdressed.
 - Know your own resume cold. **Anything on it is fair game** — if you listed a technology you touched once, be ready to say "I've used it at a project level."
-

8. If you get stuck in the room

- **Don't freeze, narrate.** "Let me think about that for a second."
- **Don't bluff.** "I haven't used that directly. My understanding is X — is that the direction you mean?" That answer has never cost anyone an offer. Bluffing has.
- **Ask for the goal.** "Is the tricky part here the state, or the API side?"
- If you fumble a question, let it go. Candidates lose interviews by mentally re-litigating question 4 while answering question 7.

10 · Talking About Your Experience

Time needed: 45 minutes. Say every answer out loud.

This file is built from your actual resume. Read it carefully, because the interviewer will have that resume in front of them and will ask about these exact lines.

First, the context: your real position in this hiring process

Read this properly, because it changes how you should behave in the room.

You are not a fresher. You have around 1 year 4 months of production experience at Zensar, building for Standard Bank South Africa, plus a 5 month internship. You have shipped features used by 10,000+ corporate banking clients.

Most people applying for this role will have college projects and a bootcamp certificate. You have:

- Real **production** React and TypeScript in a bank
- Real **performance numbers** (Lighthouse 62 → 88)
- Real **full stack** work (Spring Boot, JPA, PostgreSQL)
- Real **process** experience (SIT, UAT, Bamboo CI/CD, Agile)

So do not interview like a fresher. Do not apologise, do not undersell, and do not say "I only did small parts". Speak like someone who has shipped software that handles other people's money.

At the same time, do not be arrogant. The formula is: **calm, specific, and always backed by a number or an outcome.**

Your six best stories

Every behavioural or technical question can be answered from one of these. Learn the shape of each one. Do not memorise word for word.

Story	Use it to answer
1. My Requests & Dashboard	performance, refactoring, Redux, "biggest impact"
2. Business Card (14 screens)	complex forms, full stack, ownership, scale
3. Recall Transactions (7 screens)	multi-step forms, accessibility, replacing a manual process
4. Report Logs (AOP audit)	clean architecture, working without breaking things
5. Notifications (RabbitMQ)	event-driven thinking, real-time, accessibility

Story 1 • — My Requests & Dashboard (your performance story)

This is your strongest story for this specific job, because their job description highlights "Making Things Fast".

The 90 second version

"Corporate users of the Service Online platform had their requests spread across four separate modules. To check the status of something, they had to know which module it lived in. We consolidated all of that into one requests page and a real-time dashboard, in React 16 with Redux.

The problem after we merged everything was performance. We had pulled four modules into one page, so the initial bundle got much bigger and the page felt slow. The Lighthouse performance score was 62.

I attacked it in three ways. First, **code splitting with `React.lazy`** so each section only downloaded when it was actually opened. Second, **memoising the Redux selectors**, because unrelated state updates were re-rendering the whole dashboard. Third, general asset cleanup.

That took Lighthouse from **62 to 88**, which was roughly a 35% faster initial load. For corporate users checking requests many times a day, that is a real difference."

The follow up questions you will get, with answers

"How did you know what to fix?"

"I measured first. Lighthouse gave the headline score, and the Chrome DevTools Performance tab showed where the time actually went. The Network tab showed the bundle size problem. I did not guess, because the obvious suspect is often not the real one."

"What is selector memoization, and why did it help?"

"In Redux, a selector is a function that pulls a slice of state out of the store. If the selector creates a new array or object every time it runs, the component sees a new reference and re-renders, even when the underlying data is identical. Memoising it with `createSelector` from Reselect means the same input gives back the same reference, so the component only re-renders when the data really changed. On a dashboard with a lot of small state updates that made a large difference."

"What is code splitting?"

"By default the build produces one big JavaScript file, so the user downloads everything even if they open one section. `React.lazy` with `Suspense` splits it per section, so the code downloads only when it is needed. The first paint gets much faster because there is less to parse."

"Why was it slow in the first place?"

Be honest and non-defensive:

"Because we merged four modules into one page without splitting the bundle. It was a predictable consequence of the consolidation. We knew we would have to deal with it, and we did."

| Story 2 • — Business Card (your ownership story)

The 90 second version

"Business Card was a **14 screen onboarding application** for corporate card requests, and I built both ends of it.

On the frontend, React with **React Hook Form** and two layers of validation: field level rules as the user typed, and a full check before submission. On the backend, **Spring Boot with JPA and PostgreSQL**, and a **Camunda BPMN workflow** for the approval chain, since a card request has to pass through several approvers in order.

Before this, card processing went through a manual branch process. Digitising it **cut processing time by around 60%.**"

Follow ups

"14 screens is a lot. How did you manage the state?"

This is the strongest technical answer you own. It is also exactly the system design question in file 08.

"All the form data lived in one place at the top, not inside each screen. If each screen owned its own state, going back a step would unmount that screen and destroy what the user had typed. Keeping the data in the parent meant each screen became a presentational component that receives values and an onChange, so navigation between steps was safe and adding a screen was straightforward."

"What is Camunda, in simple terms?"

"It is a workflow engine. You draw the approval process as a BPMN diagram, with the steps and the decision points, and the engine tracks where each request currently is. The advantage is that the business rules for who approves what live in the workflow rather than being hard coded in Java, so changing the approval chain does not mean changing application code."

"What is dual layer validation?"

"Field level validation as the user fills the form, so they get feedback immediately, and a full validation before submit so nothing invalid reaches the backend. And the backend validates again independently, because frontend validation is for user experience, not for security. Anyone can bypass it."

Story 3 — Recall Transactions (your accessibility and UX story)

"Recall Transactions let corporate clients recall a payment online. Before this it could only be done by visiting a branch.

It was a **7 screen workflow** with **navigation guards**, so a user could not skip ahead past an incomplete step or lose their work by leaving accidentally. The backend was Spring Boot REST APIs with JPA and PostgreSQL, again with a Camunda process flow.

Because it was a banking flow, the forms had to be properly accessible: correct labels, keyboard navigation, and errors announced to screen readers."

"What is a navigation guard?"

"A check that runs before a route change. If the current step is incomplete or has unsaved changes, it either blocks the navigation or asks the user to confirm. In a payment recall flow, silently losing a half-completed form would be a serious problem."

Story 4 — Report Logs (your clean architecture story)

Use this when they ask about writing maintainable code.

"We needed an audit trail of who downloaded or emailed which report, which is a compliance requirement in banking. The obvious approach is to add logging code inside every download and email method, but that means touching a lot of working business logic and repeating the same lines everywhere.

Instead I used **AOP with @Aspect pointcuts**. The aspect intercepts those API calls from outside and writes the audit record, so **no existing business logic changed at all**. It persists over **50,000 transactions a day** to PostgreSQL, and we generated more than a thousand monthly reports with JasperReports."

Why this story works: it shows you think about *where* code should live, not just whether it runs. Say the line "no existing business logic changed" clearly. That is the point of the story.

Story 5 — Notifications (your event-driven story)

"Notifications were previously done by polling, so the frontend kept asking the server whether anything had happened. That wastes requests and still leaves a delay.

I replaced it with an event-driven system over **RabbitMQ**. When something happens, a message is published, and the notification is pushed instead of discovered. I also built role based notification preferences, and made the interface **WCAG 2.1 compliant**."

"Polling vs event-driven, in one line?"

"Polling is the client asking 'anything new?' over and over. Event-driven is the server saying 'here is something new' when it actually happens. Fewer requests, and no delay."

Story 6 ● — The document upload bug (your ownership story)

Use this for **"tell me about a bug you found"**, **"a time you improved someone else's code"**, **"unclear requirements"**, or **"something you fixed that nobody asked you to"**. For a plain *"hardest technical challenge"* in a frontend interview, lead with Story 1 instead, because it has a number attached.

Three things to cut from how you currently tell it

1. **Never say "being a fresher"**. You are describing yourself as junior right before describing something you did well. Delete the phrase.
2. **Never say "I solved 90% of it myself"**. It sounds like bargaining for credit and invites "so who did the other 10%?". Either you fixed it, or you fixed most of it and someone helped with a named piece. Both are fine. Vagueness is not.
3. **Do not say "it became hectic"**. Say "the harder part was". You want to sound like someone who found it interesting, not someone who found it stressful.

Also, do not spend half the answer explaining what FileNet is. The interviewer does not care about FileNet. They care what was wrong, how you found it, what you decided, and what happened.

The 90 second version

"The document upload on the card application. The business rule was that an uploaded PDF had to be within a defined size range, and the existing code only checked the upper limit. Someone had left the lower bound out.

That sounds minor, but the effect was not. A near-empty or corrupt file would pass validation, get stored in FileNet, and we would send the backend a document ID pointing at something unusable. It surfaced much later, downstream, when the document was retrieved for processing — so it looked like a backend problem when it was actually accepted at the front door.

I added the missing lower bound and validated the file type properly rather than trusting the extension. The harder part was the document ID. The file goes to FileNet, and the ID has to be stored on our side so the backend can retrieve it later. If one of those succeeds and the other fails, you get either an orphaned document or a reference pointing at nothing. So I made sure the ID was only persisted once the upload had actually succeeded.

[YOUR RESULT HERE.] What I took from it is that validation gaps do not fail where you wrote them. They fail three systems downstream, and by then nobody is looking at the upload code."

That last sentence is what makes the story land. It is the "what did you learn" that most candidates skip.

● Three things to fill in before the interview

1. The result. Right now this story has no outcome. Did invalid uploads stop? Did it get caught in SIT before release? Did support tickets drop? Anything concrete. Without a result it is a story about work, not about impact.

2. What the help actually was, if any. "I paired with a senior on the FileNet retrieval contract, since I had not worked with it before" is a mature thing to say. Naming it is fine. Gesturing vaguely at it is not.

3. Check your own numbers. A **maximum** of 3 MB is normal. A **minimum** of 2 MB is unusual — most systems set a floor of a few KB to catch blank scans. Get the real rule right, and be ready to say **why** a lower bound exists: a 1 KB PDF is an empty or corrupt scan, and it is worthless to whoever processes the application. "Why would you require a file to be at least 2 MB?" is exactly the kind of thing an interviewer at a bank-adjacent company will poke at.

Follow ups you will get

"Did you validate on the client or the server?"

The correct answer is **both**, and you should say why:

"Client side size checks are for user experience. Anyone can bypass the browser and post straight to the API, so the server has to enforce the same rule. Client side validation is never a security control."

If you only did it client side, say so honestly and add that the server side check is what you would add now. That is a better answer than pretending.

"How do you check the file type?"

"The extension and the MIME type can both be faked, so the reliable check is the file's magic bytes. A real PDF starts with `%PDF-`."

You do not need to have done this. Knowing it is the point.

"What if the upload to FileNet succeeds but saving the ID fails?"

This is the interesting one, and it is a distributed consistency problem:

"You cannot have a single transaction across two systems, so you end up with an orphaned document. The usual handling is a retry, or a cleanup job that finds documents in FileNet with no matching reference on our side."

Showing you understand the *shape* of that problem is worth more than a perfect answer.

| The QKart project

Mention it briefly if they ask about personal projects.

"QKart is a full stack e-commerce single page app I built with React, TypeScript, Redux and Material UI. Authentication, product discovery, cart and checkout over REST APIs. I cut re-renders by about 40% using custom hooks, Context API and memoisation."

If they ask **how you measured the 40%**, answer honestly:

"With the React DevTools Profiler. I recorded an interaction, looked at which components were re-rendering and why, and compared the render counts before and after."

⚠ **Make sure the repository and the live demo actually work tonight.** If an interviewer clicks the link and it 404s, that is worse than not listing it.

| • "Tell me about yourself" — your opening 90 seconds

This is the only answer worth almost memorising. Structure: **Now** → **What you built** → **Why this role**.

"I am a software engineer with about a year and a half of production experience. I was at Zensar working for Standard Bank South Africa, on their Service Online platform, which is used by over ten thousand corporate clients.

I worked across the stack. On the frontend, React and TypeScript with Redux. I shipped five features there, including a consolidated requests dashboard where I took the Lighthouse score from 62 to 88, and two end-to-end self-service applications, one of them fourteen screens, where I built both the React frontend and the Spring Boot backend.

What I enjoyed most was the frontend side, especially the performance work, and that is the direction I want to specialise in. That is why this role interested me. WeMakeScholars reaches students through search, so page speed and server rendering genuinely affect whether a student finds you and completes an application. That is the same kind of problem I was solving at Standard Bank, just with a much more direct impact."

Practise this twice tonight and twice tomorrow morning. Time it. Keep it under two minutes.

• The four hard questions your resume creates

Your resume is strong, but it raises four obvious questions. Prepare all four. Do not let any of them catch you cold.

1. "Your last role ended in March. What have you been doing since?"

There is a gap of a few months. **They will ask.** The rule: answer briefly, without apologising, and move forward.

Use whichever of these is **actually true for you**, and do not invent anything:

"I was on the bench at the end of the project and my role was released in March. I used the time to move back to Hyderabad and to go deeper on the frontend side, specifically Next.js and TypeScript, because that is the direction I want my career to go. I have been interviewing selectively rather than taking the first thing available."

The three rules for this answer:

1. **Keep it short.** Two or three sentences. A long explanation sounds defensive.
2. **Say what you did with the time.** Learning, projects, family reasons, health, relocation — whatever is true. The worst answer is "just looking".
3. **Do not apologise.** A gap after a project ends is completely normal in service companies. Treat it as normal and they will too.

Then stop talking. Do not fill the silence.

2. "Why are you leaving, or why did you leave, Zensar?"

Never criticise your employer or your client. That is the trap.

"The Standard Bank engagement was a good place to learn, and I got real ownership there. Two reasons for moving. First, I want to specialise in frontend rather than splitting across frontend and backend, and I want to work with Next.js properly. Second, I am based in Hyderabad now and I want to build here rather than in Pune. A product company also appeals to me more than services, because I would own a feature over time instead of handing it over at the end of a contract."

3. "You have Java and Spring Boot. Why do you want a pure frontend role?"

"I enjoyed both, but the frontend is where I did my best work and where I care about the details. The performance work on the dashboard was the most satisfying thing I did. Having the backend experience actually helps me here — I can read an API, tell whether a response shape will be awkward for the UI, and have a useful conversation with the backend team instead of just consuming whatever I am given."

This is a strength, not a weakness. Say it as one.

4. "How much Next.js have you actually used?"

Be precise and honest. Your resume says you *contributed to* a React 18 / Next.js / TypeScript migration. That is not the same as owning a Next.js app, and an experienced interviewer will find the difference in two questions.

"I contributed to the React 18, Next.js and TypeScript migration on the platform rather than starting a Next.js project from scratch. So I have worked in a Next.js codebase, and I have built with it outside work. I understand the rendering strategies and when to use each, but I would not claim I have run a large Next.js app in production on my own."

Then immediately show the understanding from file 03. Explain SSG, SSR, ISR and CSR and which one you would pick for their bank listing pages. Interviewers respect someone who marks the edge of their knowledge accurately, because it means they can trust everything inside it.

Questions about your numbers

Any number on your resume is fair game. Know where each one came from.

Number	Be ready to explain
Lighthouse 62 → 88	Code splitting, <code>React.lazy</code> , selector memoization
~35% faster initial load	Derived from the load time improvement, measured in Lighthouse

Number	Be ready to explain
60% faster card processing	Manual branch process replaced with a digital Camunda workflow
50,000+ daily transactions	Audit records written by the AOP aspect
10,000+ corporate clients	The user base of the Service Online platform
Core Web Vitals 65 → 82	Internship: lazy loading and asset optimisation
~40% fewer re-renders	QKart, measured with the React DevTools Profiler

⚠ If you cannot explain how a number was measured, **do not repeat it unprompted**.
A number you cannot defend does more damage than no number at all.

✓ Before you finish this file

1. Say "Tell me about yourself" out loud, twice. Time it.
2. Say the Dashboard performance story out loud, in 90 seconds.
3. Say your answer for the gap since March. Keep it under three sentences.
4. Check that your QKart repo and live demo both load.
5. Have your GitHub and QKart open in browser tabs tomorrow, ready to show.

11 · TypeScript, State Management and Testing

Time needed: 45 minutes. Read this straight after the React file.

These are all on your resume, so they are fair game. TypeScript in particular: you list it in your headline, so expect questions. Part C on choosing between Redux, Zustand and React Query is the one most likely to come up in a design or architecture conversation.

Part A · TypeScript

| First, the context: what problem does TypeScript solve?

JavaScript only tells you about a mistake when the code runs.

```
const user = { name: 'Sujith' };
console.log(user.nmae);    // undefined. No error. Silent bug.

function total(a, b) { return a + b; }
total('5', 3);            // "53". No error. Wrong answer.
```

Both bugs reach production and show up as a strange value on someone's screen.

TypeScript adds **types**, which are checked while you write the code. Both of the above become errors in your editor, before the code ever runs.

```
const user: { name: string } = { name: 'Sujith' };
console.log(user.nmae);    // ❌ Property 'nmae' does not exist

function total(a: number, b: number) { return a + b; }
total('5', 3);            // ❌ Argument of type 'string' is not assignable
```

The important detail: TypeScript disappears at build time

The browser cannot run TypeScript. It is compiled to plain JavaScript, and all the types are removed. **There is no type checking at runtime.**

This matters: if an API returns something different from what you declared, TypeScript will not catch it. It trusted you. That is why you still validate data coming from outside.

Say this

"TypeScript catches type errors while I write the code rather than when a user hits them. On a banking platform that mattered a lot, because a wrong shape in a payment form is an expensive bug. It also makes refactoring much safer, since renaming a field shows every place that breaks. The thing to remember is that the types are erased at build time, so it does not validate API responses at runtime."

Q1. `interface` VS `type` •

Both describe the shape of an object. They overlap a lot.

```
interface User {
  id: number;
  name: string;
  email?: string;           // optional
  readonly createdAt: Date;
}

type User = {
  id: number;
  name: string;
};
```

The differences that matter

1. `interface` can be reopened. `type` cannot.

```
interface User { id: number; }
interface User { name: string; }  // ✓ merged into one
```

2. `type` can do things `interface` cannot:

```
type Status = 'idle' | 'loading' | 'error';  // union
type ID = string | number;
type Point = [number, number];               // tuple
```

What to say

"I use `interface` for object shapes, especially props and API models, because it reads well and can be extended. I use `type` when I need a union, a tuple, or something computed. In practice the team convention matters more than the difference."

Q2. Typing React components • (most likely practical question)

```
interface LoanCardProps {
  bank: string;
  rate: number;
  featured?: boolean;           // optional
  onApply: (id: string) => void; // a function prop
  children: React.ReactNode;    // anything renderable
}

function LoanCard({ bank, rate, featured = false, onApply, children }: LoanCardProps) {
  return <div>{bank} - {rate}%</div>;
}
```

Typing hooks

```
const [count, setCount] = useState(0);           // inferred as number
const [user, setUser] = useState<User | null>(null); // must be explicit
const [loans, setLoans] = useState<Loan[]>([]);   // otherwise it is never[]

const inputRef = useRef<HTMLInputElement>(null);
```

"`useState(0)` infers `number` on its own. But `useState(null)` infers `null`, so if it will later hold a user I write `useState<User | null>(null)`. Same with an empty array, which would otherwise be `never[]`."

Typing events

```
const onChange = (e: React.ChangeEvent<HTMLInputElement>) => setQuery(e.target.value);
const onSubmit = (e: React.FormEvent<HTMLFormElement>) => { e.preventDefault(); };
const onClick = (e: React.MouseEvent<HTMLButtonElement>) => { ... };
```

Q3. The types worth knowing

Union — one of several options.

```
type Status = 'idle' | 'loading' | 'success' | 'error';
```

Very useful for the four UI states from file 06. The compiler then forces you to handle each case.

Generic — a type that takes a type as a parameter.

```
function first<T>(items: T[]): T | undefined {
  return items[0];
}
first([1, 2, 3]);           // returns number | undefined
first(['a', 'b']);          // returns string | undefined
```

Read `<T>` as "whatever type goes in, the same type comes out". Without generics you would write this function once per type, or fall back to `any` and lose the safety.

Utility types — built in helpers.

```
Partial<User>           // every field optional. Good for updates.
Required<User>          // every field required
Pick<User, 'id' | 'name'> // only these fields
Omit<User, 'password'>   // everything except these
Record<string, number>  // an object with string keys and number values
```

unknown vs **any** — a good question to get right.

"`any` switches type checking off completely. `unknown` also means 'I do not know the type', but it forces you to check before using it. So `unknown` is the safe version. I use `any` only as a temporary escape hatch, never as a habit."

Enum — a fixed set of named values.

```
enum Role { Admin = 'ADMIN', User = 'USER' }
```

Many teams prefer a union of string literals instead, because it produces no extra JavaScript.

Q4. Quick TypeScript questions

- **? vs | undefined?** `email?: string` means the key may be absent. `email: string | undefined` means the key must exist but may hold `undefined`.
- **What is type inference?** TypeScript works out the type without you writing it. `const n = 5` is already `number`. Do not annotate what is obvious.
- **What is `as`?** A type assertion. You are telling the compiler "trust me, this is a User". It does no checking, so it is a small risk each time.
- **Is TypeScript slower at runtime?** No. It is erased at build time. The output is plain JavaScript.
- **What is `strict` mode?** A `tsconfig` setting that turns on the important checks, most usefully `strictNullChecks`, which forces you to handle `null` and `undefined`. Worth having on.

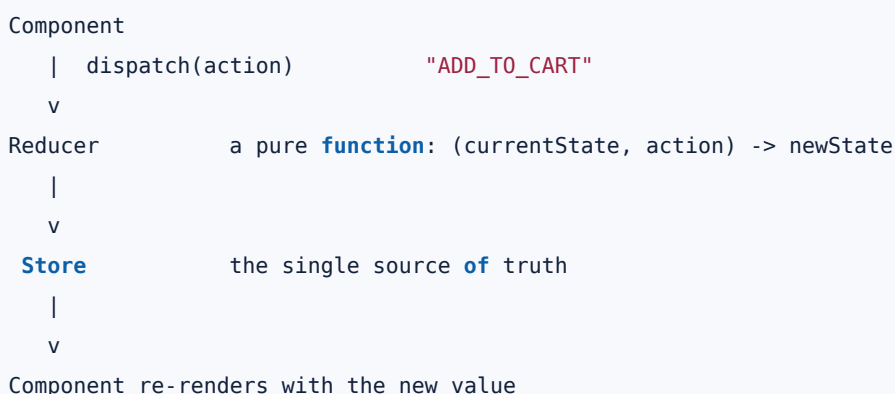
Part B · Redux and Redux Toolkit

| First, the context: what problem does Redux solve?

In React, state flows down through props. That works well until many unrelated components across the app need the same data, and events from anywhere need to change it.

Redux puts that shared state in one **store** outside the component tree. Any component can read from it, and any component can dispatch an action to change it.

The one-way cycle, which is the whole idea



The value of this shape is **predictability**. State can only change by dispatching an action, and a reducer is a pure function, so given the same state and action you always get the same result. That is why the Redux DevTools can show you every action and let you step backwards through them.

The three principles

1. **One store** for the application.
2. **State is read only.** You never mutate it directly, you dispatch an action.
3. **Reducers are pure functions.** No API calls, no random values, no mutation.

| Q5. Redux Toolkit — why it replaced classic Redux ●

Old Redux needed a lot of files: action type constants, action creators, a reducer with a switch statement, and manual immutable updates with spread everywhere.

Redux Toolkit (RTK) is the official modern way and removes most of it.

```
import { createSlice } from '@reduxjs/toolkit';

const cartSlice = createSlice({
  name: 'cart',
  initialState: { items: [] as Item[], total: 0 },
  reducers: {
    addItem(state, action) {
      state.items.push(action.payload); // looks like mutation, but it is safe
      state.total += action.payload.price;
    },
    clearCart(state) {
      state.items = [];
      state.total = 0;
    },
  },
});

export const { addItem, clearCart } = cartSlice.actions;
export default cartSlice.reducer;
```

● "That looks like you are mutating state. Is that not forbidden?"

This is the exact follow up. The answer:

"RTK uses **Immer** underneath. You write what looks like mutation, but Immer tracks the changes against a draft and produces a new immutable state object. So the rules are unchanged, the code is just far more readable than nested spreads."

Using it in a component

```
const items = useSelector((state: RootState) => state.cart.items);
const dispatch = useDispatch();
dispatch(addItem(loan));
```

Q6. Selectors and memoization ● (your dashboard story)

A **selector** is a function that reads a slice of state.

```
const selectItems = (state: RootState) => state.cart.items;
```

The performance problem you fixed at Standard Bank

```
// ❌ creates a NEW array every time it runs
const selectExpensive = (state) => state.cart.items.filter(i => i.price > 1000);
```

`useSelector` compares the result with the previous one by reference. A new array is always a new reference, so the component re-renders on **every** store update, even one that has nothing to do with the cart.

```
// <span class="tick">✓</span> memoised with Reselect (built into RTK)
import { createSelector } from '@reduxjs/toolkit';

const selectExpensive = createSelector(
  [selectItems],
  (items) => items.filter(i => i.price > 1000)
);
```

Now the filter only runs when `items` actually changes, and the same reference comes back otherwise. **This is exactly what you did to help get Lighthouse from 62 to 88.**

Q7. Async in Redux

```
export const fetchLoans = createAsyncThunk('loans/fetch', async () => {
  const res = await fetch('/api/loans');
  return res.json();
});

const loansSlice = createSlice({
  name: 'loans',
  initialState: { data: [], status: 'idle' },
  reducers: {},
  extraReducers: (builder) => {
    builder
      .addCase(fetchLoans.pending, (s) => { s.status = 'loading'; })
      .addCase(fetchLoans.fulfilled, (s, a) => { s.status = 'success'; s.data = a.payload; })
      .addCase(fetchLoans.rejected, (s, a) => { s.status = 'error'; s.error = a.error.message; });
  },
});
```

Note the three cases: pending, fulfilled and rejected. That is the loading, success and error triad from file 06, built into the tool.

● "When would you NOT use Redux?"

An important question, and the honest answer scores well.

"Most of the time. If state belongs to one component, `useState` is correct. If two siblings need it, I lift it up. If it is server data, React Query or SWR is a better fit than Redux, because caching and refetching come for free. I would reach for Redux when there is genuinely complex client state shared widely, with a lot of interactions changing it. On the Standard Bank dashboard that was justified. For a small marketing site it would be over-engineering."

Part C · Choosing between Redux, Zustand and React Query ●●

First, the context: there are two completely different kinds of state

This is the idea everything else in this section rests on. Get this right and the rest of the answers fall out of it.

Server state is data that really lives on a server. Your app only holds a copy.

- It can become **stale** the moment you fetch it, because someone else can change it.
- Getting it is **asynchronous**, so it always has loading and error states.
- The same data is often needed by several screens.
- Examples: the list of banks, a student's application status, the current user record.

Client state is data that only exists in the browser. You own it completely.

- It is **synchronous**. You set it, it is set.
- It can never be stale, because there is no other source of truth.
- Examples: is the sidebar open, which tab is selected, the current step of a form, a dark mode toggle, items in a cart before checkout.

Why this matters

Most "we need a global state manager" problems are actually server state being managed by hand. Teams put API responses in Redux, then write loading flags, error flags, refetch logic and cache invalidation themselves. That is a large amount of code that a data fetching library already gives you.

Once you separate the two, the tool choice becomes obvious:

Server state → React Query (or RTK Query, or SWR)
Client state → `useState` → Context → Zustand → Redux Toolkit

● **This is your best answer to almost any state management question.** Say the distinction first, then pick the tool. It shows you are reasoning, not reciting library names.

Q8. The decision path — what would you actually choose? ●

Work down this list. Stop at the first one that fits.

1. Only one component needs it → `useState`.

Most state is like this. Do not overthink it.

2. Two or three nearby components need it → lift it to the shared parent.

3. It comes from an API → React Query.

Caching, background refetching, deduplication, retries and loading and error states, all handled. This covers far more cases than people expect.

4. Client state, shared widely, changes rarely → Context.

Theme, language, the current user object. Context is fine here because it rarely updates, so the re-render cost does not matter.

5. Client state, shared widely, changes often → Zustand.

A cart, a complex filter panel, a wizard's state, notification toasts. Zustand gives you selector based subscriptions, so only the components reading that specific value re-render.

6. Large app, large team, complex flows → Redux Toolkit.

When you need strict conventions everyone follows, middleware, and serious debugging tools. This is where your Standard Bank platform sat.

Say this

"I start with local state and only move up when there is a real reason. The first question I ask is whether it is server state or client state, because those need different tools. Server data goes in React Query so I am not hand-writing caching and loading flags. For client state I use Context when it barely changes, Zustand when it changes often, and Redux Toolkit when the app is big enough that the conventions and the DevTools are worth the setup."

Q9. React Query — what problem does it solve? ●

The code it replaces

This is the `useEffect` pattern from file 06:

```

const [data, setData] = useState([]);
const [loading, setLoading] = useState(true);
const [error, setError] = useState(null);

useEffect(() => {
  const controller = new AbortController();
  (async () => {
    try {
      setLoading(true);
      const res = await fetch('/api/loans', { signal: controller.signal });
      if (!res.ok) throw new Error(res.status);
      setData(await res.json());
    } catch (e) {
      if (e.name !== 'AbortError') setError(e.message);
    } finally {
      setLoading(false);
    }
  })();
  return () => controller.abort();
}, []);

```

Here is the same thing with React Query:

```

const { data, isLoading, error } = useQuery({
  queryKey: ['loans'],
  queryFn: getLoans,
});

```

What you get that the hand-written version does not have

- **Caching.** Navigate away and back, the data is shown instantly from cache while it refetches quietly in the background.
- **Deduplication.** Three components asking for `['loans']` at the same time cause **one** network request, not three.
- **Background refetching.** Refetches when the window regains focus, when the network reconnects, or on an interval you choose.
- **Race condition handling.** The problem from file 06 is handled for you.
- **Retries** with backoff on failure.
- **Pagination and infinite scroll** helpers.
- **Optimistic updates** with automatic rollback.

"The point is not that it is shorter. It is that the hand-written version is only correct for the simple case. Once you need caching, deduplication and refetch on focus, you are rebuilding React Query badly."

The concepts they will ask about

queryKey — the cache key. React Query stores results against it. Include every value the query depends on, so changing a filter fetches fresh data:

```
useQuery({ queryKey: ['loans', bankType, page], queryFn: () => getLoans(bankType, page) });
```

staleTime vs **gcTime** ● — the most common React Query interview question.

What it means	
staleTime	How long the data is treated as fresh . While fresh, React Query will not refetch it. Default is 0, meaning immediately stale.
gcTime	How long unused data stays in the cache before being thrown away. Default 5 minutes. <i>(This was called cacheTime before v5.)</i>

" **staleTime** controls refetching. **gcTime** controls memory. Data can be stale but still cached, which is what makes stale-while-revalidate work: you see the old data instantly, and it updates when the fresh response arrives."

Mutations — for writes.

```
const queryClient = useQueryClient();

const { mutate, isPending } = useMutation({
  mutationFn: createApplication,
  onSuccess: () => {
    // the list is now out of date, so mark it stale and refetch
    queryClient.invalidateQueries({ queryKey: ['applications'] });
  },
});
```

invalidateQueries — "this cached data is now wrong, refetch it." This is how you keep a list in sync after creating or deleting something.

Optimistic update — update the UI before the server confirms, and roll back if it fails. Good for a like button or a checkbox, where waiting feels slow.

Q10. Zustand — what is it and why do people like it? ●

The store is just a hook

```
import { create } from 'zustand';

const useCartStore = create((set) => ({
  items: [],
  addItem: (item) => set((state) => ({ items: [...state.items, item] })),
  clear: () => set({ items: [] }),
}));
```

Using it in a component:

```
function CartCount() {
  const count = useCartStore((state) => state.items.length);
  return <span>{count}</span>;
}
```

The three things that make it different

1. No Provider. The store lives outside React. You do not wrap your app in anything. You can even read and write it from outside a component.

2. Selector based subscriptions — this is the important one. ●

`useCartStore(state => state.items.length)` subscribes **only** to that value. If some other part of the store changes, this component does not re-render.

Compare that with Context, where **every** consumer re-renders whenever the context value changes. Zustand solves the exact problem Context has.

3. Very small. Around 1 KB, against roughly 12 to 13 KB for Redux Toolkit.

Middleware worth naming

```
const useStore = create(persist(devtools((set) => ({ ... })), { name: 'cart' }));
```

`persist` saves to localStorage, `devtools` connects to the Redux DevTools extension, `immer` lets you write mutating-looking updates.

Say this

"Zustand is a minimal client state store. The store is a hook, there is no provider, and you subscribe with a selector so only the components using that specific value re-render. That last part is what makes it better than Context for state that changes often. It is a good middle ground: more structured than passing props around, far less setup than Redux."

Q11. Redux Toolkit — when is it still the right choice?

Redux is not obsolete. It is just no longer the default.

Choose Redux Toolkit when:

- The app is large and several developers work on it, and you want one enforced way of doing things.
- State transitions are complex and many parts of the app react to the same events.
- You want the **Redux DevTools**: every action logged, state inspected at each step, and time travel debugging. On a banking platform where you must explain how the app reached a state, this genuinely matters.
- You need middleware for cross-cutting concerns like logging, analytics or complex async orchestration.
- The codebase already uses it. Rewriting working state management is rarely worth it.

RTK Query is Redux's own answer to React Query, built into Redux Toolkit. If you are already on Redux, use RTK Query rather than adding React Query as a second library. They do the same job.

Q12. The comparison table •

	Redux Toolkit	Zustand	React Query
Kind of state	Client	Client	Server
Size (approx)	12–13 KB	~1 KB	~13 KB
Provider needed	Yes	No	Yes (<code>QueryClientProvider</code>)
Boilerplate	Moderate (slices)	Very little	Very little
Caching	No, you write it	No, you write it	Yes, built in
Loading / error states	You write them	You write them	Given to you
Avoids re-renders via	<code>useSelector</code> + memoised selectors	selector in the hook	per-query subscription
DevTools	Excellent	Via middleware	Excellent
Learning curve	Medium	Low	Medium
Best for	Large apps, strict conventions, complex flows	Shared client state that changes often	Anything from an API

The single most important row is the first one. Redux and Zustand solve the same problem. React Query solves a different one.

Q13. The questions you should expect, with short answers ●

"Redux or React Query — which would you pick?"

A trick question. They are not alternatives.

"They solve different problems, so it is not really either-or. React Query manages server state, Redux manages client state. A common setup is both: React Query for everything from the API, and a small Redux or Zustand store for genuine UI state."

"So does React Query replace Redux?" ● (a very common follow up)

No, and answering this precisely is worth doing. It replaces what many teams were **using** Redux for, which is not the same thing.

A typical store before:

```
// group 1 - not really state. This is a cache of what the server has.
loans: { data: [], loading: false, error: null }
banks: { data: [], loading: false, error: null }

// group 2 - genuine client state. Nothing on the server knows about this.
ui:      { sidebarOpen: false, activeTab: 'requests' }
wizard:  { step: 3, formData: { ... } }
```

React Query takes over **group 1 only**. Group 2 still needs somewhere to live: Redux, Zustand, Context, or plain `useState`.

"React Query does not replace Redux, it replaces what a lot of teams were using Redux for. Server data belongs in a cache that handles refetching and invalidation, not in a client store where you hand-write loading flags.

The reason people say it replaced Redux is that in a typical CRUD app most of the store was server data. Once that moves out, what is left is often small enough that you do not need a store at all. But that is an outcome for those apps, not a rule. Anything with genuinely complex client state, like a multi step wizard or an editor with undo, still wants a real store."

"Then why not use React Query as the store for everything?" ●

A fair question, because the React Query cache **is** global and reachable anywhere through `queryClient`. So it does look store-shaped. You can force it:

```
queryClient.setQueryData(['ui'], { sidebarOpen: true });
```

It fights the design in four ways, and each one causes real bugs.

1. Garbage collection deletes it. `gcTime` defaults to 5 minutes. Once no mounted component uses a key, the entry is dropped. That is correct for a cache, because

anything lost can be refetched. Client state has nothing to refetch from, so your wizard progress silently disappears. You would need `gcTime: Infinity` to stop it.

2. You would switch off most of the library. `staleTime`, `refetchOnWindowFocus`, `refetchOnReconnect` and retries all exist to ask "has the server changed?" None of that applies to client state. Disabling most of a library's features to use it means it is the wrong library.

3. A query with no `queryFn` is a contradiction. You end up with `enabled: false` and workarounds, and the next developer cannot tell which keys are server data.

4. No transactions, and no record of how you got here. This is the important one.

Complex state is not the same as complex data fetching

React Query is very good at complex **async** problems: caching, deduplication, invalidation, retries. What makes **client** state complex is different. It is **transitions and invariants**.

One user action often has to change several things together. In a reducer that is one action, one place describing the transition, applied atomically. With several `setQueryData` calls there is no atomicity, so a failure halfway leaves you in an invalid state. And the Redux DevTools action log answers the question you actually have when debugging complex state: *how did we get here?* The React Query DevTools show cache entries, not transitions.

For genuinely complex client state the right tools model transitions: `useReducer`, Redux Toolkit, Zustand, or a state machine such as XState. A 7 step workflow with validation between steps, navigation guards and resumability is a transition problem, not a caching problem.

The honest nuance

In many apps, what feels like complex client state is really **derived** from server state. A filter panel is the classic case: put the filters in the URL, key the query on them with `['loans', filters]`, and you genuinely do not need a store at all. That is a good architecture, and it is why the "we removed Redux" stories are real.

"You can put client state in the React Query cache, but it fights the design. The cache is garbage collected after `gcTime`, because it assumes anything lost can be refetched, and client state cannot be. You would also be turning off refetching, staleness and retries, which is most of what the library does.

The deeper reason is that complex client state is complex because of transitions: one action changing several things atomically, with a record of how you got there. That is a reducer's job. React Query is a cache with very good async handling, not a state machine."

"Why not just `useEffect` and `useState` for fetching?"

"It works for a single simple screen. It stops being enough once you need caching between screens, deduplication when several components ask for the same data, refetching when the user comes back to the tab, retries, and correct handling of race conditions. At that point you are writing a worse version of React Query."

"Why not put everything in Context?"

"Two reasons. Every consumer re-renders when the context value changes, so it is a poor fit for state that changes often. And Context is only a transport mechanism. It does not cache, refetch, or give you loading and error states, so for server data you still have to write all of that yourself."

"Redux or Zustand for a new project?"

"For most new projects, Zustand. It is far less setup and the selector based subscriptions handle re-renders well. I would pick Redux Toolkit when the app is large and several people work on it and the enforced conventions and DevTools are worth the extra weight, or when the codebase already uses it."

"What is `staleTime`?" → How long data counts as fresh before React Query will refetch it. Different from `gcTime`, which is how long unused data stays in memory.

"How does React Query know when to refetch?" → When the data is stale **and** one of the triggers fires: the component mounts, the window regains focus, the network reconnects, or you call `invalidateQueries`.

"How do you update a list after creating an item?" → A mutation, then `invalidateQueries` on the list's key in `onSuccess`. Or write the new item straight into the cache with `setQueryData` if you want to avoid the refetch.

"How does Zustand avoid re-renders without a Provider?" → The store is outside React. Each component subscribes through a selector, and only re-renders when the value that selector returns actually changes.

"Is Redux dead?"

"No, but it is no longer the automatic choice. Redux Toolkit fixed most of the boilerplate complaints, and RTK Query covers data fetching. It is still the right answer for large applications, and a lot of production code runs on it. What changed is that reaching for it on day one of a small project is now over-engineering."

• Your own answer — use your real experience

This is the strongest version of this answer available to you, because it is true:

"On the Standard Bank platform we used Redux, and later Redux Toolkit. Looking back, a lot of what we kept in the store was server data, and that meant we hand-wrote loading flags, error flags and cache invalidation for each feature.

If I were building it again, I would put the server data in React Query or RTK Query and keep the Redux store for genuine client state only. The store would be much smaller, and a whole category of bugs about stale data would just not exist.

On QKart I used Context and custom hooks with memoisation instead, which cut re-renders by about 40%. That was the right size of solution for that app — adding Redux there would have been over-engineering."

This answer does three things at once: it shows real production experience, it shows you can critique your own past decisions, and it shows you match the tool to the size of the problem. That is a senior sounding answer at any level.

Part D · Testing

First, the context: what are you actually testing?

The goal is not "100% coverage". The goal is **confidence that a change did not break something**.

Type	What it checks	Tool
Unit	One function or component alone	Jest, JUnit
Integration	A few pieces working together	React Testing Library
End to end	The whole app in a real browser	Cypress, Playwright

Q14. React Testing Library — the core idea ●

"React Testing Library's principle is to test the component the way a **user** sees it, not the way it is built internally. So I query by visible text, labels and roles, and I assert on what appears on screen. I avoid testing internal state, because if I refactor how the state is stored but the behaviour is the same, the test should still pass. A test that breaks on a refactor is a cost, not a safety net."

```
import { render, screen } from '@testing-library/react';
import userEvent from '@testing-library/user-event';

test('shows an error when the email is invalid', async () => {
  render(<LeadForm />);

  await userEvent.type(screen.getByLabelText(/email/i), 'not-an-email');
  await userEvent.click(screen.getByRole('button', { name: /submit/i }));

  expect(await screen.findByText(/enter a valid email/i)).toBeInTheDocument();
});
```

The query priority, worth knowing

`getByRole` → `getByLabelText` → `getByText` → and `getByTestId` only as a last resort. The order is deliberate: the top ones are how a real user, and a screen reader, find things.

`getBy` VS `queryBy` VS `findBy`

- `getBy` — expects it to be there now. Throws if missing.
- `queryBy` — returns `null` if missing. Use this to assert something is **not** there.
- `findBy` — waits for it to appear. Use for anything async.

Q15. Mocking an API call

```
global.fetch = jest.fn(() =>
  Promise.resolve({ ok: true, json: () => Promise.resolve([ { id: 1, bank: 'SBI' } ]) })
);
```

"I mock the network so tests are fast and do not depend on a real server being up. In a larger project I would use MSW, which intercepts at the network level, so the component code does not have to know it is being tested."

What would you test first?

A good answer, and it shows judgement:

"The parts where a bug is expensive. On the Business Card application that was the form validation and the step navigation, because a broken step could lose a user's data. I would not write tests for a component that only renders a heading."

✓ Check yourself before moving on

1. Explain in one sentence what TypeScript catches, and when the types disappear.
2. Type a React component's props, and `useState` for a value that starts as `null`.
3. Explain why RTK lets you write `state.items.push(...)` safely.
4. Explain the selector memoization problem, using your dashboard story.
5. **Explain the difference between server state and client state, then map each to a tool.** This is the highest value answer in this file.
6. Explain `staleTime` vs `gcTime`.
7. Explain how Zustand avoids re-renders without a Provider.
8. Give your own answer about what you would change on the Standard Bank store.
9. Explain the React Testing Library principle in one sentence.

12 · Final Cheat Sheet — TOMORROW MORNING ONLY

07:20 - 08:00. Nothing else. No new material.

If a line here doesn't ring a bell, open that file for 3 minutes and come back.

| The 8 sentences that carry this interview

1. **Why Next.js:** *"Plain React ships an empty HTML shell that the browser fills in — bad for first paint and unreliable for crawlers. Next renders on the server, so the browser and Google both get real HTML immediately."*
 2. **Rendering strategies:** *"SSG at build, SSR per request, ISR is SSG that regenerates on a timer, CSR in the browser. Bank listing pages → ISR. A student's own application page → SSR or CSR behind auth."*
 3. **Server vs Client Components:** *"Server by default, ships zero JS; push `'use client'` down to the interactive leaves."*
 4. **Core Web Vitals:** *"LCP under 2.5 s — is it there. CLS under 0.1 — does it stay still. INP under 200 ms — does it respond."*
 5. **Closure:** *"A function that remembers the scope it was created in. It's how `useState` and `debounce` both work."*
 6. **Keys:** *"Stable identity across renders. Index breaks the moment you delete or reorder — React reuses the wrong DOM node."*
 7. **Four UI states:** *"Loading, error, empty, success. Empty is the one people forget."*
 8. **Business tie-in (say once):** *"Since students find you through Google, server rendering and load speed here are lead generation, not polish."*
-

| Your position — remember this

- **Not a fresher.** ~1 yr 4 mo production React + TypeScript at Standard Bank via Zensar. 10,000+ corporate users. Speak accordingly.
- **Immediate joiner.** Say it clearly — most candidates are on 60-90 days.
- **Target the top of the 4-6 band.** Earn it in the tech round, then say *"given the production experience, I'd be looking at the upper end."*
- **Lead with the performance story:** Lighthouse **62 → 88**, via code splitting (`React.lazy`) + **selector memoization** (`createSelector`). Measured first.
- **Your numbers:** 14-screen Business Card (60% faster processing) · 7-screen Recall Transactions · AOP audit, 50,000+/day · RabbitMQ notifications · QKart —40% re-renders.

The 4 questions your resume creates — have all 4 ready

1. **Gap since March?** Short, no apology, say what you did with the time. Then stop.
 2. **Why leave Zensar?** Specialise in frontend · Hyderabad · product over services. Never criticise them.
 3. **Java + Spring Boot, why frontend?** It's a strength — you can read an API and tell if the shape will hurt the UI.
 4. **How much Next.js really?** "Contributed to the migration, not owned a Next app." Then immediately prove the reasoning with SSG/SSR/ISR.
-

TypeScript & Redux

- TS catches type errors **at build time**; types are **erased** at runtime, so it does **not** validate API responses.
 - `interface` for object shapes/props · `type` for unions, tuples.
 - `useState<User | null>(null)` and `useState<Loan[]>([])` — otherwise inferred as `null` and `never[]`.
 - `unknown` = safe `any`; forces a check before use.
 - **RTK + Immer:** `state.items.push()` looks like mutation but produces new immutable state. That's the follow-up they ask.
 - **Selector memoization:** a selector that `.filter()`s returns a **new array** time → new reference → re-render on every store update. `createSelector` fixes it. *(This is your dashboard story.)*
 - **When NOT Redux:** local state → `useState`; server data → React Query.
 - **RTL principle:** test what the user sees, not internal state. `getByRole` first.
-

State management — server vs client •

The distinction is the answer to almost every question here.

- **Server state** = a copy of data that lives on a server. Can go stale, is async, needs caching → **React Query** (or RTK Query).
- **Client state** = only exists in the browser. Never stale, synchronous → `useState` → Context → **Zustand** → **Redux Toolkit**.

The decision path: one component → `useState` · a few nearby → lift up · from an API → **React Query** · shared, rarely changes → Context · shared, changes often → **Zustand** · large app + team + complex flows → **Redux Toolkit**.

- **"Redux or React Query?"** is a trick question. Different problems, often both.
- **React Query gives you:** caching · deduplication · refetch on focus · retries · race conditions handled · loading and error states.
- `staleTime` = how long data counts as fresh (controls refetching).
`gcTime` = how long unused data stays in memory (controls memory). Was

`cacheTime` before v5.

- `invalidateQueries` after a mutation = "this list is now wrong, refetch it".
- **Zustand:** no Provider · store is a hook · `useStore(s => s.count)` subscribes to one value, so only that component re-renders · ~1 KB.
- **Context's weakness:** every consumer re-renders on any change, and it does no caching. Zustand fixes the first, React Query the second.
- **Your answer:** "At Standard Bank we kept server data in Redux and hand-wrote loading flags and cache invalidation. I'd put that in React Query now and keep the store for real client state. On QKart, Context plus memoisation was the right size — Redux there would have been over-engineering."

JavaScript

- `var` function-scoped · `let` / `const` block-scoped + TDZ · `const` ≠ immutable object
- Event loop: **microtasks (Promises) before macrotasks (setTimeout)** → 1 4 3 2
- `Promise.all` parallel · `allSettled` all results · `race` first settled · `any` first success
- Arrow functions take `this` from the enclosing scope and can't be rebound
- `??` only falls back on null/undefined; `||` falls back on any falsy (`0` !)
- Spread = **shallow** copy · deep = `structuredClone`
- `fetch` **does not throw on 404/500** — check `res.ok`
- Debounce = after they stop (search) · Throttle = at most once per N ms (scroll)
- `typeof null === 'object'` · `NaN !== NaN` · `Array.isArray()`

React

- Re-render causes: own state · props · **parent re-rendered** · context value
- Reduce: `React.memo` → `useCallback` / `useMemo` → move state down → `children`
- Never mutate state — new reference or no re-render (`Object.is`)
- `setCount(c => c + 1)` when the new value depends on the old
- `useEffect` = sync with the outside world; cleanup `return` ; don't use it for derived data
- `useMemo` value · `useCallback` function · `useRef` persists without re-rendering
- Hooks: top level only — React tracks them **by call order**
- Context re-renders every consumer; memoize the provider value
- `React.lazy` + `Suspense` = code splitting

Next.js

- `app/page.js` routing · `[id]` dynamic · `layout.js` · `loading.js` · `error.js`
- `fetch(url, { next: { revalidate: 60 } })` = ISR · `{ cache: 'no-store' }` = SSR
- `next/image` : WebP/AVIF, srcset, lazy, reserves space (CLS) · `priority` for the LCP image
- `metadata` / `generateMetadata` for per-page title & description

- Hydration = server HTML + client listeners; mismatches come from `Date`, `random`, `window`
- `<Link>` client-side nav + prefetch; `<a>` = full reload
- App Router → `next/navigation`; Pages Router → `next/router`
- Only `NEXT_PUBLIC_*` reaches the browser — never a secret

CSS

- `box-sizing: border-box` · Flex = 1D, Grid = 2D
- `grid-template-columns: repeat(auto-fit, minmax(260px, 1fr))` = responsive, no media queries
- Mobile-first with `min-width` · viewport meta tag or nothing works
- `display:grid; place-items:center` to centre
- Animate `transform` / `opacity` only — they skip layout and paint

APIs

- 401 = who are you · 403 = I know, still no · 429 = rate limited
- `AbortController` in the effect cleanup → no stale state, no race condition
- CORS is a browser rule, fixed on the **backend**

System design — the 6 step framework

Clarify → Components → State → Data → Edge cases → Performance

- **Never start without asking 2 questions.** "Client side or API filtering?" "How many items?" "Does this need to rank on Google?"
- **Autocomplete:** debounce 400ms · four states · cancel the old request with `AbortController` · cache per query.
- **Listing page:** keep filters **in the URL** so links are shareable, back works, and the server can render it. Reset to page 1 when a filter changes.
- **Multi step form:** all data lives in the **parent**, steps are display only. Otherwise going back destroys what was typed.
- **Reusable component:** props say *what* not *how* · use `children` · spread `...rest` · a component with 15 boolean flags should be split.

Machine coding — the 6 things they're grading

1. `key={item.id}`, never the index
2. Derived data computed during render, not stored in state + effect
3. Empty state handled ("No results for...")
4. Loading **and** error states, ideally with a retry

5. Cleanup — `clearTimeout` / `clearInterval` / `abort`

6. Talking while you type

| Opening & closing lines

"Tell me about yourself" → Present → Path → **Why here** (file 09 §1). Under 2 min.

Your closing question → *"What's the biggest frontend problem you're hoping this hire helps solve?"*

Filter questions — answer instantly, no hesitation:

18-month commitment ✓ · WFO ✓ · 1st & 3rd Saturdays ✓ · 10–11 AM login ✓

Salary → *"I saw the 4–6 LPA band and I'm comfortable in it; I'd expect to be placed based on how the technical rounds go."* **Never bid below 4.**

| The last 5 minutes before you walk in

- Close every tab except your GitHub and your two running projects.
- You know more than you think you do. Nobody expects perfection at this level — they expect someone who reasons clearly and doesn't bluff.
- **If you don't know something: say so, then reason out loud.** That answer has never lost anyone an offer. Bluffing has.
- Sit up. Breathe out slowly. Smile before you speak — it changes your voice.

Go get it.